Why Does CBT Fail? A Critical Review of Ten Proposed Explanations, with Executive Function and Adult ADHD as a Test Case

Brando Miranda*

Stanford University, Stanford, CA, USA

Abstract

Non-response to cognitive behavioral therapy (CBT) does not identify a single failure of treatment, theory or delivery. Evidence that a treatment package works does not establish which components cause improvement or why some patients do not benefit. We critically evaluate ten proposed reasons CBT may fail, using selected meta-analyses, component trials and mechanistic studies rather than claiming systematic coverage. The review covers efficacy, mechanisms, patient and context factors, and delivery. Executive function and adult attention-deficit/hyperactivity disorder provide the deepest case study. CBT remains effective on average, but benefits depend on the comparator, and many limitations are shared with other treatments. Component findings are mixed and often underpowered; they establish neither the universal necessity nor the irrelevance of cognitive restructuring. For example, a depression review (16 studies, 22 comparisons) favoured full treatments over versions with one component removed ($g = 0.21$, 95% confidence interval $[0.03, 0.39]$), despite substantial methodological limitations. Evolutionary conservation motivates learning-based hypotheses but cannot rank therapies or establish that experience must precede cognition. Researcher allegiance and publication selection warrant scrutiny without establishing CBT-specific reviewer suppression. Acceptance, alliance, reactance and feedback research further qualify accounts that isolate technique from goals and delivery. We propose falsifiable component and moderator tests of which interventions produce durable functional change, for whom and under what conditions.

*Corresponding author

Email address: brando9@stanford.edu (Brando Miranda)

Draft 4.1: for feedback, not submitted.

Keywords: cognitive behavioral therapy, cognitive restructuring, executive function, anxiety, ADHD, non-response, dropout, relapse, therapist effects, critical review

1. Introduction

Cognitive behavioral therapy (CBT) is among the most heavily tested psychotherapies. A meta-analysis of 41 placebo-controlled trials in anxiety and related disorders (total $N = 2{,}843$) found that CBT outperformed placebo on target symptoms with a moderate effect (Hedges' $g = 0.56$) (Carpenter et al., 2018). The same literature also shows a ceiling: across 87 studies only about half of patients respond to CBT for anxiety disorders, and the figure depends on how response is defined (Loerinc et al., 2015); a larger meta-analysis of 100 articles reports a mean remission rate of 51.0% (Springer et al., 2018). A therapy can work on average and still leave open *why* it works and *for whom*.

The textbook account of CBT's cognitive component is a top-down loop. A situation triggers an automatic thought ("I am going to fail this talk"); the patient records it, names the distortion, weighs evidence for and against, and generates a more balanced thought; the feeling is expected to follow. This review first maps ten reasons given for why CBT fails (Section 3) and then grades the evidence for each. Its deepest case study is a narrow question about that loop: *does the mechanism ask the patient to use a capacity that anxiety, and in some patients attention-deficit/hyperactivity disorder (ADHD), has already degraded?* This motivates tests of whether executive capacity and stress moderate cognitive skill use, and whether components with different executive demands produce different outcomes. Executive demand alone does not establish those clinical differences.

The argument has a popular version. Judson Brewer, a clinician-researcher, argues that anxiety takes the prefrontal cortex "off-line," so that willpower and reasoning are unavailable at the moments when they are needed (Brewer, 2016, 2021), and he has applied this directly to CBT (Brewer, 2026). Our aim is to find out how much of this survives contact with the primary literature. Our answer, in brief, is that a narrow version survives and the strong version does not. The same standard is applied to the other reasons: dropout and non-response (Section 12), patient and context factors (Section 13), therapist and delivery factors (Section 14), and the theory, measurement and evidence-quality critiques (Section 15).

What we find. (1) Reappraisal does draw on working memory in laboratory studies, but evidence that acute stress abolishes it is thin and contested. (2) The direct evidence that executive function moderates CBT outcome is sparse and mostly null for overall outcome, with the clearest signal at the level of the cognitive skill itself. (3) Component findings are mixed: several trials detect no added benefit, whereas a depression component review favours full packages and some direct comparisons favour cognitive therapy; many designs are underpowered. (4) CBT's headline effects shrink with active controls and bias adjustment; whether they shrink over time is disputed. (5) In adult ADHD, CBT is modestly effective, mostly as an adjunct to medication, and the plausible active ingredient is behavioural compensation; we found no trial testing ADHD as a moderator of CBT for anxiety in adults. (6) Beyond that argument, dropout (about 16% before and 26% during CBT), non-response, relapse and under-recorded harm are real and quantified, but mostly shared with other psychological treatments; patient, context and delivery factors shift outcomes without isolating a CBT-specific failure; and mechanism, measurement and trial-quality limitations constrain causal interpretation and effect estimates.

Contributions. (i) An explicit decomposition of cognitive restructuring into executive demands (Section 4); (ii) an evidence-graded review of the stress-and-prefrontal premise, including a check of the "off-line" claim against its sources (Section 5); (iii) a review of efficacy, evidence selection and component evidence (Sections 7–8); (iv) a review of alternatives and of CBT in adult ADHD (Sections 9–10); (v) the direct moderator evidence (Section 11); (vi) a graded map of ten reasons CBT may fail (Section 3) and reviews of non-response, patient, delivery and mechanism factors (Sections 12–15); (vii) an analysis of evolutionary learning and clinical goals and engagement, with explicit limits on the inferences they support (Sections 15.1 and 13.1); (viii) testable hypotheses (Section 17).

2. Method

This is a critical theoretical review, not a systematic review reported in full under the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. We select literatures relevant to ten reasons CBT may fail, with executive function and adult ADHD as the deepest case study. The initial searches ran in October 2026 across PubMed, Europe PMC,

Crossref, publisher pages and reference lists of key meta-analyses. They covered efficacy and bias; components and mechanisms; executive function and stress; ADHD; non-response, dropout, relapse and harm; patient and context factors; delivery; and theory and measurement. We prioritised meta-analyses, component trials, moderator studies, and critiques and defences of CBT. A targeted extension in November 2026 examined evolutionary learning, researcher allegiance and publication selection, acceptance, engagement, alliance and intervention timing. This extension followed proposed claims and relevant counterevidence; it was not an exhaustive update of every literature.

Verification. For the extension, we checked the title, authors and publication year of scientific references against DOI-resolved Crossref records. We distinguished this bibliographic check from reading abstracts or relevant full-text passages to verify findings. Quoted numbers came from those passages, not from search summaries or memory. The public source notes record reading levels, scope limits, inaccessible candidates and retrieval provenance; copyrighted reading copies remain outside the public repository. Earlier source notes and the public verification log document prior DOI checks and abstract reading; their partial-verification flags remain in place. Popular sources motivate questions but do not establish empirical findings.

Limitations of the method. Selection was not independently duplicated, the review was not preregistered, and we did not formally assess the risk of bias of every included review. Our synthesis therefore remains vulnerable to selection and interpretation bias, including our own preference for the proposed argument. Evidence from depression is used where anxiety evidence is thin, with the distinction stated. A structured umbrella search with independent screening and risk-of-bias assessment remains necessary before claiming systematic coverage (Section 17).

3. A map of the reasons CBT fails

"CBT does not work" is not one claim. It bundles at least ten different ones, which differ in what they say fails (the average effect, the individual response, the mechanism, the evidence or the delivery), in how well they are supported, and in whether they are specific to CBT at all. Table 1 lists them with the verdict this review reaches and the section that holds the evidence. We use four verdicts: **supported** (consistent evidence, quantified), **partly supported** (real but smaller or more conditional than the popular version),

contested (credible evidence on both sides) and **open** (little or no direct evidence).

Two cautions apply throughout. First, many of the best-documented limits (dropout, relapse, comorbidity, harm under-reporting, therapist variability) are limits of psychological treatment in general, not of CBT in particular; we say so each time the source does. Second, a limit on certainty is not evidence of ineffectiveness: CBT keeps a moderate average effect under almost every sensitivity analysis we found (Plessen et al., 2023; Carpenter et al., 2018). The map is therefore a list of *where and why* CBT falls short of its reputation, not a verdict that it fails.

The executive-function argument (R1) is the review's deepest case study, because it makes a testable, mechanism-level prediction and because it was the starting point of the project. It is one reason among ten, and Sections 12–15 treat the others at the same standard of evidence.

4. What cognitive restructuring asks of the patient

Consider the standard thought-record exercise. The patient must (a) *notice* an automatic thought while it is occurring, (b) hold the thought, the supporting evidence, the contradicting evidence and a candidate alternative in mind at once, (c) suppress the pull of the automatic thought while evaluating it, and (d) shift from being inside the feeling to observing it. Mapped onto the best-known model of executive function, these are three separable components: updating and holding information (working memory), inhibiting a prepotent response, and shifting between mental sets (Miyake et al., 2000).

Laboratory evidence that reappraisal draws on executive resources. In event-related-potential studies, high working-memory load eliminated the effect of reappraisal on the late positive potential (Gan et al., 2017; Adamczyk et al., 2022). Working-memory capacity predicted reappraisal success (Schmeichel et al., 2008), and reappraising late in an emotional response, once emotion is already intense, costs self-control resources (Sheppes and Meiran, 2008). People also choose differently as intensity rises: they prefer distraction to reappraisal in high-intensity negative situations and reappraisal in low-intensity ones (Sheppes et al., 2011).

Evidence that it works, and can be trained, when load is low. Instructed reappraisal reduces negative experience in the laboratory (Hofmann et al., 2009) and can be trained (Denny and Ochsner, 2014). So the claim is not

	Reason	Verdict	Where
R1	The cognitive skill demands executive function that anxiety (and ADHD) degrade	contested; direct moderator evidence open	Sections 4–6, 11
R2	The incremental benefit of cognitive restructuring is inconsistent	contested; depends on disorder and comparison	Section 8
R3	Headline effects are inflated by waitlists, bias, low-quality trials and measurement choices	supported	Sections 7, 15
R4	Many patients do not respond or drop out	supported (not CBT-specific)	Section 12
R5	Gains erode after treatment ends	partly supported	Section 12
R6	Harms and deterioration are under-recorded	supported for reporting; CBT lowers deterioration versus controls	Section 12
R7	Patient factors (personality disorder, chronicity, adversity, medical comorbidity, neurodivergence)	partly supported; mostly not CBT-specific; autism evidence thin	Section 13
R8	Context: WEIRD trials, culture, waiting times, deprivation, access	supported	Section 13
R9	Delivery: therapist variability, drift, weak training, dose, format	partly supported; some causal evidence	Section 14
R10	The theory of change is under-tested; common factors may explain more	contested	Section 15

Table 1: **CBT's limitations concern different outcomes and levels of explanation.** Ten proposed reasons, with the verdict of this critical review.

that reappraisal fails; it is that it *costs* executive resources, which matters when those resources are scarce.

Two clarifications. First, this is a task analysis plus laboratory support, not a treatment finding. Clinicians scaffold the load (worksheets, a therapist who prompts for evidence, graded practice). Second, executive function is not a single resource; the argument concerns working memory, inhibition and shifting, not general intelligence. We return to both in Section 16.

5. Stress, anxiety and the prefrontal executive system

5.1. Mechanistic neuroscience

The most developed account is due to Arnsten. Her review of stress signalling describes the prefrontal cortex (PFC) as the most evolved brain region and also the one most sensitive to the detrimental effects of stress exposure (Arnsten, 2009). The proposed mechanism is an inverted-U in catecholamine signalling: moderate norepinephrine and dopamine levels support prefrontal network firing and working memory, while the high levels of uncontrollable stress weaken them and strengthen amygdala and striatal habit circuitry. A later review summarises the result as stress rapidly flipping the brain from reflective to reflexive control of behaviour (Arnsten, 2015). Acute uncontrollable stress impairs PFC-dependent working memory in monkeys and rodents (Arnsten and Goldman-Rakic, 1998). In humans, a meta-analysis finds that acute stress impairs working memory and cognitive flexibility, with more nuanced effects on inhibition and many moderators (Shields et al., 2016), and imaging suggests a reallocation of resources toward salience networks rather than a switch-off (Hermans et al., 2014).

Where the "off-line" language comes from. The phrase traces to a hedged line in a 1998 monkey study ("Stress may take the PFC 'off-line"'; Arnsten and Goldman-Rakic 1998), and its author still uses it in quotation marks for *uncontrollable* stress (Arnsten, 2020). Brewer and Roy apply it to worry and cite Arnsten's 2009 and 2015 reviews, which are not studies of worry (Brewer and Roy, 2021); their app-based trial ($n = 65$, treatment-as-usual control) did not test a prefrontal mechanism (Roy et al., 2021). We therefore read "offline" as a popularisation of *impaired, in a graded and dose-dependent way*, mainly under uncontrollable stress.

5.2. Does stress abolish reappraisal in humans?

Evidence for this specific link is contested. The founding study found that participants stressed before learning showed no fear reduction after regulation training (Raio et al., 2013), and a review summarises that acute stress may impair the persistent inhibition of fear, presumably by altering prefrontal function (Raio and Phelps, 2015). Five later studies ($n = 54$–98) found no group-level impairment (Shermohammed et al., 2017; Sandner et al., 2021; Langer et al., 2021, 2023; Hamza et al., 2024), though two reported sex- or cortisol-linked effects. Their laboratory stressors were moderate, so the null results may not transfer to clinical-level arousal; that gap is itself an open question.

5.3. Cognitive psychology of anxiety

Attentional Control Theory proposes that anxiety impairs the inhibition and shifting functions of executive control, so that anxious individuals are more captured by threat and worse at redirecting attention (Eysenck et al., 2007). The theory predicts that *efficiency* (effort and time) suffers more than *effectiveness* (accuracy), because anxious people compensate with extra effort. Meta-analytic support exists at group level ($g = -0.58$; Shi et al., 2019) and for working memory ($g = -0.334$; Moran, 2016), but with qualifiers: a meta-analysis of 32 studies ($N = 13{,}084$) found generalised anxiety disorder (GAD) associated with poorer cognitive flexibility and working memory but not inhibitory control, and with no association with any executive domain in older adults (Nguyen et al., 2025). Working memory was unrelated to social anxiety in one study (Harrison et al., 2025), and one study found *enhanced* control over negative distractors in GAD (Hallion et al., 2019).

6. Executive function in ADHD

ADHD is heterogeneous: executive difficulties are common but not universal. A meta-analysis of 83 studies found that groups with ADHD underperform controls on every executive task class examined, most strongly on response inhibition, vigilance, working memory and planning, with medium effect sizes (0.46–0.69) (Willcutt et al., 2005). The same authors caution that executive weaknesses are neither necessary nor sufficient to cause all cases of ADHD, and fewer than half of the (mostly child) samples show impairment on any single executive test (Willcutt et al., 2005). Measured more carefully, impairment is more prevalent: in a sample of 136 children, 89% of those with ADHD

were impaired on at least one executive function (62% working memory, 27% inhibitory control, 38% set shifting) (Kofler et al., 2019).

Anxiety is common in adult ADHD: a meta-analysis of 58 studies (N = 18,821) found higher anxiety scores in adults with ADHD than without ($g = 0.77$; $g = 0.95$ against non-clinical and $g = 0.44$ against clinical controls) (Hoskins et al., 2026). Whether anxiety adds executive load in ADHD is less clear: in a sample of 339 children, trait anxiety did not uniquely predict working memory or inhibitory control, nor moderate the ADHD effect (Marsh et al., 2024). The stacking assumption (low baseline executive capacity, lowered further by anxiety) is therefore plausible but, in children at least, not supported by the one careful test we found.

7. What the efficacy evidence shows

7.1. The case for CBT

Against placebo, CBT outperforms on anxiety target symptoms ($g = 0.56$ across 41 trials; Carpenter et al. 2018). Against inactive controls, effects are larger: a meta-analysis of 144 trials found $g = 0.75$ (depression) to 0.88 (social anxiety disorder), with 0.80 for GAD and 0.81 for panic disorder (Cuijpers et al., 2016). A 2025 network of 375 trials gave anxiety-spectrum effects between 0.5 and 1.0, above 1.0 for PTSD and specific phobia (Cuijpers et al., 2025). In social anxiety, individual CBT ranked best of the therapies compared in a network meta-analysis of 101 trials (Mayo-Wilson et al., 2014), and in GAD only CBT remained significantly better than treatment as usual at 3–12 months in a network of 65 trials (Papola et al., 2024). Any critique has to explain why CBT works at all.

7.2. The headline shrinks with better controls and bias adjustment

- **Control type.** More than 80% of anxiety trials used waitlist controls, and the few that did not showed much smaller effects (Cuijpers et al., 2016); only 17.4% of trials in that analysis were high quality. In the 375-trial analysis, waitlist-controlled effects exceeded $g = 0.94$, while care-as-usual comparisons gave 0.22–1.13 (Cuijpers et al., 2025).

- **Bias.** In depression, psychotherapy effects fell from $g = 0.70$ to $g = 0.31$ after adjusting for several sources of bias, with only 23% of studies at low risk of bias (Cuijpers et al., 2019b). In a cohort of National Institutes of Health-funded depression trials, 13 of 55 initiated grants did not lead to

publication (Driessen et al., 2015). Adding six unpublished studies to 20 published control comparisons changed the pooled effect from $g = 0.52$ (95% CI $[0.37, 0.68]$) to 0.39 ($[0.08, 0.70]$). The reported 25% decrease is a change in point estimates; the published–unpublished contrast was uncertain ($p = .07$), and the pooled treatment effect remained positive. An umbrella review of 102 recent meta-analyses (3,782 RCTs) found most target-symptom effects small and pooled psychotherapy effects of 0.34 against placebo or treatment as usual (Leichsenring et al., 2022).

- **Response and remission.** About half of anxiety patients respond or remit (Loerinc et al., 2015; Springer et al., 2018).

7.3. Evidence volume, allegiance and reviewer selection

Evidence volume, comparative efficacy and component causality are different questions. A large literature can increase confidence in a treatment, but its size alone establishes neither superiority to active alternatives nor which component causes improvement. Researcher allegiance—investigators' preference for a treatment—was associated with comparative outcomes across 30 meta-analyses (Munder et al., 2013). That association warrants scrutiny, but does not alone separate bias from differences in treatment efficacy, expertise or implementation. Nor does allegiance adjustment necessarily eliminate a CBT advantage: one review retained an advantage after adjustment, with superiority over psychodynamic but not interpersonal or supportive therapies (Tolin, 2010). These findings justify examining particular comparisons, not declaring all therapies equivalent.

Publication bias must also be distinguished from reviewer preference. Among the 13 unpublished studies identified by Driessen et al. (2015), two were known to have been submitted and rejected, six were never submitted, and submission status was unclear for five. This is not evidence that CBT-oriented reviewers excluded competing therapies. A randomized experiment did find that peer reviewers favoured a positive-result manuscript over an otherwise comparable no-difference version, but it involved orthopaedic journals, not psychotherapy (Emerson et al., 2010). We did not identify direct evidence establishing CBT-specific reviewer suppression in the sources examined for this extension. Testing that hypothesis requires reviewer-orientation data and comparable manuscripts, with non-submission, editorial decisions and review recommendations distinguished. Prospective registration, complete reporting, credible active comparators and independent replication address

evidence selection without presuming that one therapeutic orientation is uniquely biased.

7.4. Does the effect shrink over time?

This is contested, and the evidence is mixed. In depression, one meta-analysis reported a steady decline in CBT effects since its introduction (Johnsen and Friborg, 2015), but a reanalysis restricted to randomised comparisons judged the decline "most likely a spurious finding" (Cristea et al., 2017), and a 409-trial meta-analysis found no reduction by publication year (Cuijpers et al., 2023). In anxiety, a meta-regression of 49 controlled studies over 30 years found a flat trend (publication year $B = -0.008$, $p = .24$): effects "did not increase over the last 3 decades" (Hofmann et al., 2025). Placebo-controlled anxiety estimates have been reported at lower values in successive reviews (0.73, 0.56, 0.24), but those come from separate reviews with different case mixes, not from one meta-regression. We therefore do *not* claim a decline over time; the defensible statement is that effects have not grown and that they are sensitive to control condition and bias.

8. Where the active ingredients may lie

8.1. Component and dismantling evidence

The foundational depression dismantling trial ($N = 150$) detected no advantage of complete cognitive treatment over behavioural activation, with or without automatic-thought skills, at termination or six months (Jacobson et al., 1996). A larger depression trial ($N = 440$) established behavioural activation's non-inferiority to CBT on the Patient Health Questionnaire-9 (PHQ-9) at 12 months (Richards et al., 2016). The modified intention-to-treat difference was 0.1 PHQ-9 points (95% CI $[-1.3, 1.5]$), within the trial's prespecified 1.9-point margin; the per-protocol analysis supported the same inference. Behavioural activation was delivered by junior mental health workers and CBT by psychological therapists, so this pragmatic trial compared treatment packages and workforces rather than isolating one component. It supports a viable treatment without an explicit restructuring module, not identical effects or a particular learning mechanism. In anxiety:

- In a 108-trial meta-analysis, cognitive therapy, exposure and their combinations were efficacious with "no differential efficacy for any treatment components" (Norton and Price, 2007).

- In a full-factorial trial of internet CBT for social anxiety ($N = 464$), adding psychoeducation or exposure improved post-treatment outcomes; adding cognitive restructuring or attention training did not show superiority (Šipka et al., 2025). The psychoeducation and exposure advantages were no longer significant four months after treatment, as conditions including cognitive restructuring or attention training improved further. Substantial attrition limits the follow-up interpretation. The exposure component itself included recording and comparing expected and observed outcomes, so it did not exclude cognitive processing.

- Adding cognitive restructuring to prolonged exposure did not enhance outcome in assault survivors with PTSD (Foa et al., 2005), nor in children with social anxiety (de Mooij et al., 2023).

- A component network meta-analysis in panic disorder (72 studies) singled out interoceptive exposure, not cognitive restructuring, as efficacy-enhancing (Pompoli et al., 2018).

- In GAD, the network meta-analysis authors concluded that pure cognitive restructuring was not a decisively active ingredient when considered alone (Papola et al., 2024); the paper was later corrected for a duplicated cohort and an error in standard-error handling (JAMA Psychiatry, 2024).

8.2. Evidence against the claim

These findings are not uniform.

- Cognitive therapy beat exposure in social phobia in a meta-analysis of 20 direct comparisons ($N = 1{,}308$), with no difference in PTSD, OCD or panic (Ougrin, 2011).

- In a trauma trial, adding cognitive restructuring to exposure was associated with PTSD rates of 31% versus 63–75% without it at 6 months (Bryant et al., 2008).

- In-session cognitive restructuring correlates with outcome ($r = .35$), though from only four studies and correlationally (Ezawa and Hollon, 2023).

- In a single-session trial with elevated public-speaking anxiety ($N = 249$), restructuring before exposure and behavioural experiments both outperformed exposure without explicit expectancy processing (Johnco et al., 2025). The small, non-significant difference in anxiety reduction favoured behavioural experiments, which also produced greater expectancy change in a shorter session. Follow-up was limited to one week.

- The exposure–cognition split is blurred: exposure may itself work by changing the expectancy of harm (Hofmann, 2008).

8.3. Why the component evidence is weak on both sides

A review of 16 depression component studies found a small pooled advantage for full treatments over versions with one component removed ($g = 0.21$, 95% CI $[0.03, 0.39]$) (Cuijpers et al., 2019a). The components differed, so this is not an estimate of cognitive restructuring's unique contribution. Only one study had low risk of bias, and all but one lacked power to detect effects smaller than $g = 0.55$. Mediation evidence is correlational: cognitive change tracks symptom change however it is produced (Cristea et al., 2015; Lorenzo-Luaces et al., 2015), cognitive mediation of symptom reduction was limited in a depression cohort (median $r = .06$; Vittengl et al., 2014), and reviewers say no common or specific factor meets causal criteria (Kazdin, 2007; Cuijpers et al., 2019c). A null component result is thus weak evidence that restructuring adds nothing, and a positive correlation is weak evidence that it matters.

8.4. The strongest defence: practice when calm

A defender can answer that patients rehearse when calm rather than undertake effortful restructuring during peak distress. We accept the force of this possibility, but successful rehearsal would not by itself identify practice or habit formation as the active ingredient. It could improve retrieval, cognitive skill, expectancies or behavioural routines; these alternatives require measurement and component tests. Repeated performance does not reliably produce outcome-insensitive habits in human laboratory tasks (de Wit et al., 2018). Exposure protocols emphasise learning through expectancy violation (Craske et al., 2014). The relevant questions are what is rehearsed, what changes, and whether the acquired response transfers to situations with high executive demands.

9. Alternatives and their evidence

9.1. Mindfulness and acceptance-based treatments

Mindfulness and acceptance-based approaches train attention and responses to thoughts and feelings rather than relying exclusively on disputing thought content. Acceptance-based work overlaps with the broader CBT family (Hayes et al., 2006). Its inclusion here does not establish an absence of cognitive or executive demands. Brewer frames anxiety as a habit loop (trigger, worry, a feeling of control) and proposes noticing, curiosity about the reward value of worry, and a "bigger, better offer" (Brewer, 2021). Evidence for the specific programme is early: in a randomised trial of an app-based programme for GAD ($n = 65$), anxiety scores fell 67% versus 14% under usual care, with mediation implicating reduced worry (Roy et al., 2021). The trial is small, self-reported, compared with usual care rather than CBT, and has the programme's developer among its authors.

The following syntheses do not establish that mindfulness and acceptance approaches outperform evidence-based comparators. A meta-analysis found essentially no difference ($g \approx 0.00$) between mindfulness-based interventions and evidence-based treatments, often CBT (Goldberg et al., 2018); another found no difference from traditional CBT or behavioural therapies ($g = -.07$, nine comparisons; Khoury et al. 2013). For acceptance and commitment therapy (ACT) versus CBT, the pooled difference was non-significant in one meta-analysis (A-Tjak et al., 2014), and a sceptical reading of 60 RCTs concluded that ACT is not yet well-established for any disorder (Öst, 2014). In 276 adults with anxiety disorders, eight weeks of mindfulness-based stress reduction was noninferior to escitalopram, with fewer discontinuations for adverse events (Hoge et al., 2023). That is a non-inferiority result against a first-line drug, not evidence of superiority to CBT. These comparisons do not establish universal equivalence across disorders, outcomes and follow-up periods. Similar outcomes can arise through shared or different processes; they do not identify a mechanism by themselves.

Reward revaluation as a hypothesis. An alternative self-regulation account proposes that attending to the experienced consequences of an action can change its reward value and reduce reliance on effortful restraint (Ludwig et al., 2020). This is a mechanistic proposal, not comparative evidence against CBT. An app-based eating study linked model-estimated reward values to reported eating, but had no control group (Taylor et al., 2021). Its larger

naturalistic sample showed decreasing estimated reward values without a corresponding decrease in reported eating behaviour across tool uses. The study supports further testing of reward revaluation, not a causal conclusion about the intervention, a general mechanism for anxiety, or superiority to cognitive restructuring.

9.2. Medication in ADHD

In a network meta-analysis of 133 double-blind trials, amphetamines (standardised effect -0.79) and methylphenidate (-0.49) outperformed placebo in adults on clinician ratings, over short follow-up (Cortese et al., 2018). The certainty varied from high or moderate to low or very low. A Cochrane review of extended-release methylphenidate found a self-rated effect of -0.37, judged 20 of 21 trials at high risk of bias, and noted that all trials excluded participants with psychiatric comorbidity such as anxiety or selected previous responders (Boesen et al., 2022). So the pivotal medication trials say little about adults with ADHD *and* anxiety.

10. CBT in adult ADHD

10.1. Efficacy

A randomised trial of 86 medication-treated adults found that CBT outperformed relaxation with educational support, with 53% versus 23% responders on clinician global ratings (Safren et al., 2010); a smaller earlier trial found 56% versus 13% responders for CBT added to medication (Safren et al., 2005). A meta-analysis of 32 studies found a between-group effect of $g = .65$ on symptoms, smaller with active controls, and neither medication status nor treatment length moderated outcome (Knouse et al., 2017). Another found SMD 0.76 against waitlist but 0.43 against active controls (Young et al., 2020), and a 2026 meta-analysis of 14 RCTs found SMD -0.45 on core symptoms and -0.43 on executive function (Liu et al., 2026). A component network meta-analysis of 113 RCTs found CBT better than placebo only on clinician-reported measures (Ostinelli et al., 2025), and blinded-rater effects of psychological treatments were attenuated to non-significance in an earlier synthesis that did not separate adult from child samples (Sonuga-Barke et al., 2013). An umbrella review judged CBT, atomoxetine and methylphenidate to have at least moderate-certainty evidence in adults, with no high-certainty long-term evidence for any intervention (Gosling et al., 2025). The Cochrane review rates the CBT evidence as low quality (Lopez et al., 2018).

10.2. What the protocols contain, and where CBT sits

The effective protocols are skills-heavy: time management, organisation and planning, with adaptive thinking as one module (Safren et al., 2010; Solanto et al., 2010). One review argued that learning behavioural compensatory strategies "may be a critical active ingredient" (Knouse and Safren, 2010), and the authors of one trial proposed behavioural control as the key mechanism (Young et al., 2017). We found no dismantling study of cognitive restructuring in adult ADHD, so this is inference. In the largest multi-centre trial ($N = 433$), group CBT did not beat structured clinical management, while methylphenidate beat placebo (Philipsen et al., 2015). Guidelines position CBT as an adjunct to medication (Kooij et al., 2019; National Institute for Health and Care Excellence, 2018). Results on adding medication to CBT are mixed: no incremental benefit in a small double-blind pilot (Weiss et al., 2012), but greater early improvement in an unblinded trial ($N = 88$), with the gap narrowing over follow-up (Cherkasova et al., 2020).

10.3. Anxiety in ADHD

In a meta-analysis of 13 RCTs ($n = 744$), CBT-based interventions reduced anxiety in ADHD modestly (SMD -0.29 post-treatment, -0.42 at follow-up), with effects more apparent in adults; most trials delivered ADHD-focused rather than anxiety-targeted CBT (Wang et al., 2026). Another meta-analysis found CBT improved anxiety, particularly at follow-up, predicted by core-symptom reduction (López-Pinar et al., 2020). In children with primary anxiety in group CBT ($N = 842$), neither ADHD diagnosis nor subtype predicted response or remission, which cuts against the strong form of our thesis (Gould et al., 2018). The adult version of that question appears untested.

11. Does executive function moderate CBT outcome?

This is the question on which the argument stands or falls. The direct evidence is thin and mostly from older adults.

- **Supports (at the level of the skill).** In older adults, poorer cognitive flexibility predicted poorer cognitive-restructuring quality and less distress reduction from restructuring, but *not* overall CBT outcome: those with poorer flexibility "can still benefit from standardised CBT" (Johnco et al., 2014).

- **Supports (pilot).** In a pilot of CBT for GAD in older adults, participants whose executive scores remained low did not respond, while those whose scores improved responded as well as an intact-executive group (Mohlman and Gorman, 2005).

- **Null or mixed.** No baseline executive effect was found in adult OCD (Vandborg et al., 2016), child anxiety (Godovich et al., 2020) or brief psychotherapy for late-life depression (Schurr et al., 2025); a systematic review of late-life anxiety found no replicated executive predictor (Kyriacou et al., 2023). In a meta-analysis of antidepressant response in late-life depression, only planning and organisation predicted outcome (Pimontel et al., 2016).

- **Opposite direction.** Better executive performance went with non-response to exposure-based CBT in paediatric OCD (Hybel et al., 2017), and worse set-shifting went with better CBT response in late-life depression (Goodkind et al., 2016).

- **Adaptation helps.** Problem-solving therapy beat supportive therapy in older adults with executive impairment, with a larger advantage at greater impairment (Areán et al., 2010), which points to adapting therapy rather than abandoning it.

Taken together, "low executive function means CBT fails" is not supported. A narrower claim is: executive capacity may limit uptake of the *cognitive skill* without limiting overall response, because CBT also works through routes that do not need it. We found no adequately powered study of working-age adults with ADHD and anxiety.

12. Non-response, dropout, relapse and harm

Dropout. Dropout is the most visible failure. A meta-analysis of 115 CBT studies (20,995 participants) found 15.9% dropout before treatment and 26.2% during it, highest in depression and in e-therapy (Fernandez et al., 2015). CBT is not unusual here: across 669 psychotherapy studies the weighted dropout was 19.7% and was not moderated by therapy orientation (Swift and Greenberg, 2012); in 54 depression trials it was 17.5% in psychotherapy arms (Cooper and Conklin, 2015), and in routine-care CBT for depression about 25% (Hans and Hiller, 2013). CBT dropout is, if anything, lower

than medication dropout (Cuijpers et al., 2013a). Reported predictors are a weak alliance ($d = .55$; Sharf et al., 2010), comorbid personality disorder and minority status (Cooper and Conklin, 2015), and, in self-guided online programmes, male sex, low education and comorbid anxiety (Karyotaki et al., 2015).

Non-response. Even under trial conditions many patients do not respond. In a moderate-to-severe depression trial, 16-week response was 58% and remission 40% for cognitive therapy, with outcomes depending on therapist experience at one site (DeRubeis et al., 2005). Across 92 psychotherapy studies fewer than half of patients met response or remission criteria, and 48% of care-as-usual patients also no longer met depression criteria, so the added value was about 14 percentage points (Cuijpers et al., 2014). For anxiety, one review puts non-response at about a third of completers and finds that "no predictor has been consistently supported" (Taylor et al., 2012); see also Loerinc et al. (2015) and Springer et al. (2018). Routine services do less well than trials: of 19,395 English IAPT patients, 40.3% reliably recovered, 63.7% reliably improved and 6.6% reliably deteriorated (Gyani et al., 2013). The prognostic factors with support (comorbid personality disorder, chronicity, adversity) apply to all treatments (Section 13); baseline severity did not moderate CBT versus medication in an individual-participant meta-analysis of 1,700 patients (Weitz et al., 2015). In fairness, CBT-based collaborative care beat usual care in primary-care anxiety (12-month response 63.7% versus 44.7%; Roy-Byrne et al., 2010) and adjunctive CBT roughly doubled response in treatment-resistant depression (46% versus 22%; Wiles et al., 2013), with a benefit still present at 3–5 years (Wiles et al., 2016); the Cochrane review rates the evidence for this as resting mostly on one trial (Ijaz et al., 2018).

Relapse and durability. CBT has an enduring effect relative to stopping medication (relapse OR 2.61 favouring CBT; no significant difference from continued medication; Cuijpers et al., 2013a; Hollon et al., 2005). Yet after acute-phase cognitive therapy ends, 29% of patients relapse within one year and 54% within two (Vittengl et al., 2007); continuation-phase CBT reduces this, especially in high-risk groups (Jarrett et al., 2001), and preventive psychological interventions cut relapse against usual care (RR 0.64; Biesheuvel-Leliefeld et al., 2015). In anxiety the follow-up record is thin: effects were not significant at 12 or more months for panic disorder, and relapse was reported in only 6 of 69 trials (van Dis et al., 2020). In a panic

trial CBT-alone response fell from 48.7% acutely to 31.9% six months after stopping (Barlow et al., 2000).

Deterioration and harm. Across guided internet-CBT trials deterioration was lower with treatment than in controls (5.8% versus 17.4%, Rozental et al., 2017; 3.36 versus 7.60, Ebert et al., 2016), but low-education patients were at higher risk and did not differ from controls in the latter. Broader unwanted effects are common and poorly recorded: 5.2% of 14,587 English therapy patients reported lasting bad effects (Crawford et al., 2016); unwanted events are expected in roughly 5–20% of patients (Linden and Schermuly-Haupt, 2014); therapists of 100 CBT patients reported side effects in 43 (Schermuly-Haupt et al., 2018); and a review of 85 trials found harmful events in more than 1 in 10 participants (Klatte et al., 2023). Only 21% of 132 psychological-intervention trials monitored harms (Jonsson et al., 2014), and none of 82 NIHR psychological trials mentioned an adverse event in the final report (Duggan et al., 2014). These are general to psychological therapy (Lilienfeld, 2007), and the CBT literature has itself called for better harm detection (Barlow, 2010). Routine outcome feedback for patients who are off track is a cheap mitigation (d = 0.23; Delgadillo et al., 2018; Lambert et al., 2018).

Reading. Dropout, non-response, relapse and harm are real, quantifiable and under-reported, so R4–R6 stand. They show that CBT leaves a large minority behind, not that it fails relative to alternatives, and almost none of the sources isolates the cognitive component.

13. Patient and context factors

Comorbidity, chronicity and adversity. Comorbid personality disorder more than doubles the odds of a poor depression outcome (OR 2.16), whatever the treatment modality (Newton-Howes et al., 2014); childhood maltreatment predicts non-response or non-remission (OR 1.43; Nanni et al., 2012); and psychotherapy for chronic depression has a small effect against controls ($d = 0.23$) and did worse than medication in direct comparisons ($d = -0.31$, attributed to dysthymia studies; Cuijpers et al., 2010b). These are prognostic factors for all treatments, not demonstrations that CBT specifically fails. For anxiety, overall diagnostic comorbidity was generally unrelated to effect size across 148 treatment samples, with diagnosis-specific exceptions (Olatunji et al., 2010). Trauma-focused CBT for PTSD in military samples shows large within-group effects (d 0.78–1.10) yet about two thirds of patients retained

the diagnosis, and the programmes were only marginally better than non-trauma-focused active controls (Steenkamp et al., 2015). Transdiagnostic CBT is one response to comorbidity (Newby et al., 2015). Baseline severity does not moderate CBT versus medication (Weitz et al., 2015), and CBT still helps medication-refractory depression (Wiles et al., 2016).

Neurodivergence, cognitive impairment, age and medical illness. The evidence on autism is thin rather than negative: ten randomized trials in autistic adults (n = 537) leave most effects very uncertain (Perez-Castilla et al., 2026), and an earlier mixed-age meta-analysis found that effects depended on who rated them ($g = 0.24$ self-report, non-significant; 0.66 informant; 0.73 clinician; Weston et al., 2016). In intellectual disability, pooled effects were small ($g = 0.43$, and 0.36 in randomized trials; Tapp et al., 2023); in dementia, CBT-family therapies had a small effect on depression (SMD −0.23) and an uncertain one on anxiety (Orgeta et al., 2022). Age itself is not a barrier: older adults recovered more often in routine IAPT data (OR 1.33; Saunders et al., 2021), and late-life anxiety CBT remitted 54% of patients, though about a third relapsed (Johnco et al., 2026). Medical comorbidity attenuates effects: CBT for chronic pain gives very small to small benefits (SMD −0.09 to −0.34; Williams et al., 2020), and insomnia CBT works but is weaker with medical than psychiatric comorbidity (Wu et al., 2015). In substance use, CBT beats minimal treatment but not other specific therapies (Magill et al., 2019). We found no meta-analysis of educational attainment or IQ as a moderator. Whether adult ADHD moderates CBT outcome is addressed in Section 11.

Culture, representation and access. Culturally adapted CBT outperforms other conditions ($g = 0.67$) and unadapted versions ($g = 0.52$; Hall et al., 2016; $d = 0.32$ in Benish et al., 2011, where adapting the explanation of the illness was the one significant moderator); depression trials show similar gains (Anik et al., 2021), and CBT transports to Chinese-descent samples with modification (Li et al., 2023). But the trial base is narrow: across 342 depression trials, linguistic minorities were far more likely to be excluded than included and effects were almost never reported by ethnic group (Polo et al., 2019), in line with the broader WEIRD-sampling critique (Henrich et al., 2010). Access shapes outcomes: in IAPT, waiting time and missed appointments predicted worse outcomes and service organisation explained 33% of the variance in reliable improvement (Clark et al., 2018); recovery

was lower in poorer areas (Delgadillo et al., 2016). In low-resource settings, lay-delivered CBT-family therapy works (adjusted mean difference −4.86 on the SSQ-14 in Zimbabwe, Chibanda et al., 2016; −7.57 on the BDI-II in India, Patel et al., 2017; OR 0.22 for major depression in Pakistan, Rahman et al., 2008), so there CBT fails through reach, not efficacy, with the pooled evidence rated mostly low certainty (van Ginneken et al., 2021; Barbui et al., 2020).

Rumination and worry. Rumination predicts the onset and course of depression and overlaps with worry (Nolen-Hoeksema et al., 2008). They are also targets: rumination-focused CBT improved residual depression in a small trial with mediation by rumination change (Watkins et al., 2011), and repetitive-negative-thinking training reduced 12-month depression and anxiety diagnoses against a waitlist (Topper et al., 2017). This suggests a process that standard restructuring may miss, not a demonstrated failure.

Reading. R7 and R8 are supported in a limited sense: patient and context factors shift outcomes, but nearly all are shared with other treatments, and the strongest CBT-specific gaps are evidence gaps (autism, cognitive impairment, minority groups) rather than measured failures.

13.1. Goals, engagement and clinical formulation

A clinically useful distinction is between reducing an unpleasant feeling and changing how a person responds to it. In ACT, acceptance and cognitive defusion—changing one's relationship to thoughts rather than treating them as literal directives—serve action guided by values (Hayes et al., 2006). The proposed target is *psychological flexibility*: persisting or changing behaviour in accordance with values and the present situation. A review of 12 mediation studies found results generally consistent with this model, but identified methodological limitations and narrow coverage of the proposed processes (Stockton et al., 2019). This is preliminary process evidence, not proof of a unique causal mechanism or superiority to CBT. It also limits a feeling-first account: valued action need not be conceptualised as waiting for distress to disappear.

A practicing clinician's perspective, shared informally with the author, distinguishes needing instructions, wanting change but being blocked by fear, resisting perceived control, and difficulty establishing collaborative engagement. We treat these as formulation questions, not a validated four-type classification, fixed traits or diagnoses. We did not identify validation of

that exact classification in our targeted searches and the readiness/reactance sources reviewed here; these searches were not exhaustive. No prevalence claims follow, and disengagement alone does not establish a psychiatric diagnosis.

That perspective also distinguishes *intellectual understanding*—recognising that a feared outcome is unlikely—from changes in emotional responding and everyday behaviour. A patient may endorse that judgement yet continue to feel threatened and avoid the situation. We treat this as a formulation question, not a prevalence estimate or proof of a separate non-cognitive system.

Related constructs do have an empirical literature. Pretreatment readiness to change was associated with psychotherapy outcome across 76 studies and 25,917 patients (Krebs et al., 2018). That association does not establish the benefit of assigning treatment by a readiness category. A synthesis of 13 controlled studies with 1,208 patients found that people high in reactance—resistance to perceived control—did better with less directive than directive treatment (Beutler et al., 2018). This supports considering the interaction between engagement and delivery, not attributing every disagreement or poor outcome to a resistant patient. For our argument, the implication is conditional: the usefulness of an instruction or cognitive technique may depend on how it is communicated and received. It does not follow that cognitive techniques are useless.

14. Therapist, delivery and real-world factors

Therapists. Therapist effects are real but modest: about 5% of outcome variance on average (range 0.2–29%; Johns et al., 2019), 6–7% among low-intensity workers in IAPT (Firth et al., 2015b), and in one large routine dataset individual therapists' recovery rates ranged from 23.5% to 95.6% (Saxon and Barkham, 2012). Whether delivering CBT competently or adherently matters is unsettled: a meta-analysis found adherence–outcome and competence–outcome correlations near zero ($r = .02$ and .07; modalities pooled; Webb et al., 2010), while single CBT studies find competence effects mostly at the extremes or in subgroups (Branson et al., 2015; Strunk et al., 2010; Kuyken and Tsivrikos, 2008). Experience alone did not improve outcomes across 6,591 patients and 170 therapists (Goldberg et al., 2016). The therapeutic alliance includes agreement on goals and tasks and a therapeutic bond. A synthesis of 295 studies found a face-to-face alliance–outcome association of $r = .278$

(95% CI [.256, .299]), with a similar association for internet therapy (Flückiger et al., 2018). This is correlational and not CBT-specific. In one sample of 331 patients seen by 80 therapists, therapist-level but not patient-level alliance variation predicted outcome (Baldwin et al., 2007). Session-by-session data from 17 studies and 5,350 participants showed reciprocal relations between early alliance and symptoms, rather than a simple one-way account (Flückiger et al., 2020).

Drift and training. Reviews argue that clinicians under-use core behavioural components and drift from "doing" to "talking" therapy (Waller, 2009; Waller and Turner, 2016); a survey found most licensed psychologists did not use exposure for PTSD (Becker et al., 2004) (dated, US). We found no pooled prevalence of drift. Training changes behaviour only when it is active and supervised: unsupervised internet training did not change competence in a randomized trial (Rakovshik et al., 2016), in line with broader reviews (Beidas and Kendall, 2010; Shafran et al., 2009).

Dose, homework and schedule. Homework compliance correlates with outcome ($r = .26$, 23 studies; Mausbach et al., 2010; $g = 0.79$ for quantity, Kazantzis et al., 2016), but patients who improve may do more homework. Improvement is curvilinear with dose, and most responders in routine care respond within 7 (low-intensity) or 14 (high-intensity) sessions (Robinson et al., 2020); session frequency may matter more than total sessions (Cuijpers et al., 2013b). Intensified exposure shortened time to response in a randomized trial; its authors reported formally equivalent symptom reduction at post-treatment and six months (Pittig et al., 2021).

Does CBT survive contact with routine care?. Largely yes. Effectiveness studies for anxiety lie within the range of efficacy trials (Stewart and Chambless, 2009), those for depression are somewhat lower with about 25% dropout (Hans and Hiller, 2013), and clinical representativeness did not erase effects (Shadish et al., 2000). IAPT shows pre–post d of 0.87 (depression) and 0.88 (anxiety) (Wakefield et al., 2021), though year-one reliable recovery was 40.3% (Gyani et al., 2013), and a small independent diagnostic follow-up found only 9.2% fully recovered (n = 90; single author; contested; Scott, 2018). Youth evidence-based practices beat usual care only modestly (mean effect 0.29; Weisz et al., 2013).

Format and personalisation. A synthesis of 20 studies with 1,418 participants found a small average difference between guided internet and face-to-face CBT ($g = 0.05$, 95% CI $[-0.09, 0.20]$), with limited evidence for individual conditions (Carlbring et al., 2018). Guided internet CBT works in routine care (Etzelmueller et al., 2020), and guided beats unguided only slightly at post-treatment (Karyotaki et al., 2021); unguided CBT has a small effect ($g = 0.27$; Karyotaki et al., 2017). A large pragmatic trial of supported computerised CBT in primary care found no benefit over usual care, with low uptake (Gilbody et al., 2015). Group and individual formats showed little average difference when treatments, patients and doses were matched (Burlingame et al., 2016). Truijens et al. (2019) found no support for manual superiority in six direct comparisons, but three of eight indirect meta-analyses favoured it. These findings do not establish general superiority of manualised care, nor does the absence of superiority establish equivalence. Algorithm-matched care modestly beat stepped care (OR 1.40; Delgadillo et al., 2022; see also Firth et al., 2015a).

Relationship, feedback and timing. Repairing alliance ruptures was associated with better outcomes across 11 studies ($r = .29$, 95% CI $[.10, .47]$), but six training/supervision comparisons did not establish an outcome benefit ($r = .11$, $[-.09, .30]$) (Eubanks et al., 2018). A cluster-randomized trial provides more direct evidence for structured feedback: among 2,233 patients, those not on track improved more when therapists received progress feedback (Delgadillo et al., 2018). This supports monitoring and adaptation, not the claim that every informal therapeutic probe is effective.

The concern that an early reframe may feel invalidating is a plausible clinical hypothesis, but the cited alliance and reactance findings do not directly test that timing mechanism. Relational readiness must also be distinguished from whether restructuring occurs before or after exposure. In 93 participants with claustrophobic fear, restructuring before exposure and consolidation after exposure produced no detected difference in outcomes through one month (Krause et al., 2022). The cognitive exercises differed as well as their order, and the trial was not an equivalence test. The public-speaking-anxiety trial supports explicit expectancy processing before or after exposure, with a small, non-significant anxiety advantage and greater expectancy change in the after-exposure condition (Section 8). Alliance, communication and timing may condition a technique's usefulness.

The informal clinical perspective proposes a sequence in which experien-

tial and emotional change precede a final phase of *cognitive consolidation*: deliberate reflection on what changed and how to sustain it. It also allows earlier reframing when that helps a person engage in a different experience. This is a proposed therapeutic sequence, not a universal rule established by the timing trials above. Consolidation after an individual exposure and consolidation near the end of therapy are different timing questions.

Reading. R9 is partly supported. CBT works in routine care with variable yield, and some delivery changes, including structured feedback, have causal evidence. How much non-response is attributable to delivery rather than treatment content or patient/context factors remains unresolved.

15. Theory, mechanism and measurement critiques

Mechanisms are under-tested. Kazdin (2009) argues that "there is no evidence-based explanation of how or why even the most well-studied interventions produce change", and sets standards (temporal precedence, specificity, dose–response, experiment; Kraemer et al., 2002) that most CBT mediation studies do not meet. Positive evidence exists: sudden symptom gains were preceded by cognitive change in two datasets (Tang and DeRubeis, 1999; Tang et al., 2005). It is mixed: dysfunctional-attitude change mediated symptom reduction in CBT versus interpersonal therapy, but the reverse direction fit CBT versus medication ($n = 130$; Quilty et al., 2008); the CBT-side reviews call the evidence "not yet conclusive" (Driessen and Hollon, 2010) and the evidence for skill use as a mediator preliminary (Hundt et al., 2013). Cognitive vulnerability–stress interactions predicted later depression only weakly in youth (Lakdawalla et al., 2007). Defences of the cognitive model and its updates are in Beck and Haigh (2014) and Hofmann et al. (2013).

Is the cognitive part the active part?. In a network meta-analysis of 45 trials ($n = 3{,}382$), cognitive restructuring, behavioural activation and full CBT did not differ, and all beat usual care (SMD 0.57, 0.52, 0.44) (Ciharova et al., 2021). This is consistent with a behavioural reading and equally with a cognitive one; see Section 8 for the dismantling trials. For exposure, inhibitory learning emphasises new learning through expectancy violation (Vervliet et al., 2013; Weisman and Rodebaugh, 2018). In this framework, new associations can compete with earlier threat learning rather than erase it (Craske et al., 2014). This gives a learning-based account of a corrective experience, not evidence that emotional intensity itself is therapeutic. It is not necessarily

an alternative to belief change: revising an expectancy can itself constitute cognitive change, even without explicit restructuring (Hofmann, 2008). A large trial found expectancy-violation-based exposure effective and optimisable (Pittig et al., 2021). Attempts to suppress unwanted thoughts can produce a later rebound (Wenzlaff and Wegner, 2000; Abramowitz et al., 2001). The meta-analysis found a small-to-moderate effect that varied with the target thought and its measurement, with few clinical samples (Abramowitz et al., 2001). This concerns suppression and should not be treated as evidence that cognitive restructuring causes rebound. Experiential avoidance gives a third-wave counter-theory (Hayes et al., 1996), though a key measure may largely track distress (Wolgast, 2014) and third-wave trials were less stringent than CBT trials (Öst, 2008). The process-based critique of one protocol per diagnosis (Hayes and Hofmann, 2021; Hofmann and Hayes, 2018) comes from advocates of the alternative.

Communication format is not a mechanism. Communication format—for example, verbal discussion versus an action-based exercise—does not identify the process responsible for change. The clinical perspective proposes that a verbal reframe can help a person experience a different response, rather than merely communicate an explanation. Conversely, behavioural experiments can change expectancies, as in the public-speaking trial (Section 8). An experience-based description does not establish an exclusively subcortical mechanism. Nor do the component and timing comparisons directly validate brief strategic therapy as a treatment package.

Common factors. The alliance and therapist effects reviewed in Section 14, together with expectancy ($r = .18$; Constantino et al., 2018), are replicable and modest. Wampold (2015) argues they outweigh specific ingredients, and Enck and Zipfel (2019) frame context effects in psychotherapy; this is contested, and it explains why trials without blinding cannot cleanly isolate CBT's specific effect.

Measurement. Clinician-rated scales show larger effects than self-report ($\Delta g = 0.20$; Cuijpers et al., 2010a). Seven depression scales share little symptom content (mean overlap 0.36; Fried, 2017); four common scales lack unidimensionality and longitudinal invariance ($n = 3{,}509$; Fried et al., 2016), and sum scores hide which symptoms change (Fried and Nesse, 2015). Response is therefore partly instrument-dependent, and standardised, patient-defined outcomes are rare (Cuijpers, 2019; Lambert and Ogles, 2009).

Evidence quality. Only 11 of 115 depression trials met eight quality criteria, with $d = 0.22$ versus 0.74 in the rest (Cuijpers et al., 2009); excess significant findings and mean power of 49% suggest small-study bias (Flint et al., 2014); only 15% of audited trials were prospectively registered (Cybulski et al., 2016); and selective reporting inflated effects (SMD -0.81 versus -0.54; Miguel et al., 2021). A Cochrane-method review rated all eight cognitive-therapy-versus-usual-care trials at high risk of bias (Jakobsen et al., 2011), though risk-of-bias tools fit psychotherapy awkwardly (Munder and Barth, 2017). Fairness requires the counterweight: a multiverse analysis of 4,281 meta-analyses still gave a mean $g = 0.56$, 90% clinically relevant, though pooling all psychotherapies (Plessen et al., 2023). Field-level critiques call for mechanism research and less redundant trials (Cuijpers, 2015; Tackett et al., 2019).

Reading. R10 and the measurement and bias part of R3 identify limits on certainty. These findings do not establish a single mechanism of CBT-specific failure. They show that causal accounts of change remain under-tested and that trial selection and measurement can inflate estimated effects.

15.1. Evolutionary learning: motivation, not a treatment ranking

The evolutionary argument begins with a defensible observation: learning from consequences does not require human verbal reasoning. Operant reward learning has been demonstrated in the mollusc *Aplysia* (Brembs et al., 2002). Comparative work also identifies conserved basal-ganglia organization across lamprey and mammals (Grillner and Robertson, 2016). Neither observation compares psychotherapies or establishes that an older mechanism produces greater clinical benefit. The prefrontal cortex is not a single recent human addition: its subdivisions include shared mammalian regions and later primate specializations (Preuss and Wise, 2022). Evolutionary conservation supplies biological plausibility, not an ordering of treatment efficacy.

The learning concepts also require separation. *Instrumental learning* concerns relations between actions and their consequences; *Pavlovian learning* concerns predictive relations between events (Dolan and Dayan, 2013). Within instrumental control, goal-directed action is sensitive to current outcome value, whereas habitual control is relatively insensitive to outcome devaluation. These are interacting forms of control, not a literal division between a cognitive cortex and a non-cognitive basal ganglia (Dolan and Dayan, 2013). Repeating an action in a stable context can support habit formation, but repetition

is not itself a separate learning system (Wood and Rünger, 2016). Five unsuccessful human habit-induction experiments caution against assuming that practice necessarily yields outcome-insensitive responding (de Wit et al., 2018). Reward-related motivation is also not identical to conscious pleasure: wanting and liking can dissociate (Berridge and Robinson, 2016).

The useful clinical hypothesis is therefore about the relation between understanding and action, not the age of a brain region. For persistent avoidance under high executive demand, structured behavioural practice may produce more durable functional change than restructuring without comparable practice. This proposed comparison concerns acquisition, generalization and cognitive cost; it is not an established treatment ranking. Nor does it require that an unpleasant feeling disappear before action changes, as the acceptance-based formulation makes explicit (Section 13.1).

The stronger claim that cognition can only consolidate experience is contradicted by laboratory evidence. Instructed reward probabilities improved decisions in one learning task (Li et al., 2011). Misleading instructions biased choices despite contrary experience in another (Doll et al., 2009). These findings show that instructions can influence learning and behaviour, not that cognitive restructuring is invariably useful in clinical care. The distinction between communication format and mechanism also permits cognitive preparation for behavioural practice. The exposure-component and timing evidence therefore constrains any universal "experience first" claim (Sections 8 and 14). The defensible research question is which combination of practice, cognitive processing and relational support works for a specified problem and patient, not which system is evolutionarily older.

16. The strongest objections

1. **CBT has a large evidence base.** Yes (Section 7). The argument concerns mechanism and moderators, not average efficacy.
2. **CBT for adult ADHD works.** It beats matched controls in medicated adults with a skills-heavy protocol; the evidence is graded low to moderate (Section 10).
3. **Patients practise when calm.** Accepted as a plausible defence (Section 8). Rehearsal could improve cognitive skill as well as behavioural routines; it does not identify habit formation as the mechanism.
4. **Exposure is behavioural.** This is why CBT cannot be equated with cognitive restructuring. The distinction between a procedure and its

mechanism matters (Section 8).

5. **Mindfulness is not better.** The comparative evidence does not establish general superiority. The hypothesis concerns when particular components help, not whether one therapy family should replace another.
6. **Cognitive therapy sometimes wins.** Yes, in social phobia (Ougrin, 2011) and in one PTSD trial (Bryant et al., 2008).
7. **The neuroscience is about acute stress.** Correct, and the human reappraisal-under-stress findings are mixed (Section 5).
8. **Executive function does not predict CBT outcome.** Largely true for overall outcome (Section 11); the argument survives only in the narrower, skill-level form.
9. **Therapists scaffold the executive load.** Plausible and untested; a therapist who generates the evidence list lowers demand and could erase a moderator effect.
10. **Evolutionary age does not rank therapies.** Correct. Conservation motivates a learning-based hypothesis but cannot establish comparative clinical benefit (Section 15.1).
11. **Cognition can help before experience.** Correct. The learning and timing evidence is reviewed in Section 15.1; acceptance-based goals are distinguished from eliminating distress in Section 13.1.
12. **Bias is not proof that efficacy is illusory.** Correct. Researcher allegiance, publication selection and reviewer preference are distinct; they apply to competing orientations too and do not establish treatment equivalence (Section 7.3).

Limitations. This is a narrative review by a non-clinician, with no independent duplicate screening. Several key studies involve older adults, children or elevated-symptom rather than diagnosed-disorder samples. Evidence from depression informs some anxiety questions, and laboratory learning is not a direct test of psychotherapy. The clinical heuristic is not a validated taxonomy, and the added literature searches were targeted rather than exhaustive. No mechanism proposed here should be treated as an established explanation of an individual patient's outcome.

17. Predictions and a research agenda

The argument yields hypotheses, not established treatment-selection rules.

1. **P1 (skill-level moderation).** Baseline executive function predicts acquisition of restructuring skill more strongly than acquisition of exposure-based skills, but predicts overall outcome only weakly.
2. **P2 (stress-dependence).** The gap between calm and in-the-moment reappraisal is larger in low-executive-function samples and at higher arousal than under the moderate stressors studied so far.
3. **P3 (component).** In adults with ADHD and anxiety, an exposure-and-behaviour protocol is non-inferior to the same protocol plus restructuring, under a clinically justified, prespecified margin.
4. **P4 (ADHD).** ADHD status predicts less benefit from the cognitive module specifically, rather than necessarily less benefit from the full CBT package.
5. **P5 (practice).** For persistent avoidance under high executive demand, structured behavioural practice produces more durable functional improvement than restructuring without comparable practice. This prediction concerns a defined behavioural target, not all distress or every patient.

A component comparison should include restructuring, behavioural practice and their combination, with comparable therapist contact and credible rationales. Measure actual practice, expectancy change, skill acquisition, executive demand and functioning rather than inferring mechanisms from treatment labels. Measure intellectual understanding, reported threat, avoidance and functioning separately, including change outside sessions. Acceptance-related change should include valued action despite distress, not only reductions in distress intensity. Greater emotional intensity during a session is not by itself evidence of durable change. To test sequencing, hold cognitive content and contact as comparable as possible, distinguish early alliance from prior symptom change, and measure engagement alongside follow-up outcomes. A final-phase cognitive consolidation hypothesis also requires outcomes beyond the final session, rather than treating a persuasive explanation as evidence of lasting change. **What would weaken the hypotheses:** a precisely estimated absence of the predicted executive-function interaction would weaken P1. A precisely estimated absence of a larger calm-versus-stressed reappraisal gap in low-executive-function samples would weaken P2. A clinically meaningful restructuring benefit in the specified population would contradict P3. A precisely estimated absence of a difference in cognitive-module benefit by ADHD status would weaken P4. A precisely estimated absence of a practice

advantage would weaken P5. A non-significant difference alone is insufficient: the relevant superiority or non-inferiority contrast, margin, outcome, follow-up and power must be specified before data collection. Mediation analyses also require temporal ordering and assumptions about confounding; correlated process change is not enough. The immediate research priorities are a structured umbrella search with independent screening, analyses of existing participant-level data where available, and adequately powered component/moderator designs. Any prospective clinical comparison would require ethical review and appropriate treatment provision; none is conducted here.

18. Conclusion

The ten reasons graded in Table 1 do not support the claim that CBT generally fails. They identify substantial non-response, relapse, under-recorded harms, comparator-sensitive effects, uncertain mechanisms, variable delivery and unequal access. Most of these limitations are shared with other psychological treatments. The executive-function argument remains narrower than its popular formulation: reappraisal uses working memory, but stress does not reliably abolish it, and clinical moderation remains insufficiently tested. Component evidence does not establish that explicit restructuring is always necessary, always dispensable, or the unique route to cognitive change. Evolutionary conservation motivates the learning-based hypothesis, whose clinical value requires direct testing. Acceptance, alliance, reactance and feedback research further caution against treating treatment content independently of goals and delivery. Researcher allegiance and publication selection warrant scrutiny, but do not establish CBT-specific reviewer suppression or invalidate the efficacy literature. The useful next question is which components, delivered in which way, help which patients achieve durable functional change.

Declarations

Funding: none. **Competing interests:** none declared. **Author note:** this is a theoretical critical review by a non-clinician; it is not medical advice. **Data and materials:** search, bibliographic and source-check records are in the public repository at `https://github.com/brando90/meta-review-cbt`. **Generative artificial intelligence (AI) assistance:** generative AI tools assisted literature retrieval, drafting and checking. Author verification is required before submission.

References

A-Tjak, J.G., Davis, M.L., Morina, N., Powers, M.B., Smits, J.A., Emmelkamp, P.M., 2014. A meta-analysis of the efficacy of acceptance and commitment therapy for clinically relevant mental and physical health problems. Psychotherapy and Psychosomatics 84, 30–36. doi:10.1159/000365764.

Abramowitz, J.S., Tolin, D.F., Street, G.P., 2001. Paradoxical effects of thought suppression: a meta-analysis of controlled studies. Clinical Psychology Review 21, 683–703. doi:10.1016/s0272-7358(00)00057-x.

Adamczyk, A.K., Wyczesany, M., van Peer, J.M., 2022. High working memory load impairs reappraisal but facilitates distraction – an event-related potential investigation. Biological Psychology 171, 108327. doi:10.1016/j.biopsycho.2022.108327.

Anik, E., West, R.M., Cardno, A.G., Mir, G., 2021. Culturally adapted psychotherapies for depressed adults: A systematic review and meta-analysis. Journal of Affective Disorders 278, 296–310. doi:10.1016/j.jad.2020.09.051.

Areán, P.A., Raue, P., Mackin, R.S., Kanellopoulos, D., McCulloch, C., Alexopoulos, G.S., 2010. Problem-solving therapy and supportive therapy in older adults with major depression and executive dysfunction. American Journal of Psychiatry 167, 1391–1398. doi:10.1176/appi.ajp.2010.09091327.

Arnsten, A.F., 2020. Guanfacine's mechanism of action in treating prefrontal cortical disorders: Successful translation across species. Neurobiology of Learning and Memory 176, 107327. doi:10.1016/j.nlm.2020.107327.

Arnsten, A.F.T., 2009. Stress signalling pathways that impair prefrontal cortex structure and function. Nature Reviews Neuroscience 10, 410–422. doi:10.1038/nrn2648.

Arnsten, A.F.T., 2015. Stress weakens prefrontal networks: molecular insults to higher cognition. Nature Neuroscience 18, 1376–1385. doi:10.1038/nn.4087.

Arnsten, A.F.T., Goldman-Rakic, P.S., 1998. Noise stress impairs prefrontal cortical cognitive function in monkeys. Archives of General Psychiatry 55. doi:10.1001/archpsyc.55.4.362.

Baldwin, S.A., Wampold, B.E., Imel, Z.E., 2007. Untangling the alliance-outcome correlation: Exploring the relative importance of therapist and patient variability in the alliance. Journal of Consulting and Clinical Psychology 75, 842–852. doi:10.1037/0022-006x.75.6.842.

Barbui, C., Purgato, M., Abdulmalik, J., Acarturk, C., Eaton, J., Gastaldon, C., Gureje, O., Hanlon, C., Jordans, M., Lund, C., Nosè, M., Ostuzzi, G., Papola, D., Tedeschi, F., Tol, W., Turrini, G., Patel, V., Thornicroft, G., 2020. Efficacy of psychosocial interventions for mental health outcomes in low-income and middle-income countries: an umbrella review. The Lancet Psychiatry 7, 162–172. doi:10.1016/s2215-0366(19)30511-5.

Barlow, D.H., 2010. Negative effects from psychological treatments: A perspective. American Psychologist 65, 13–20. doi:10.1037/a0015643.

Barlow, D.H., Gorman, J.M., Shear, M.K., Woods, S.W., 2000. Cognitive-behavioral therapy, imipramine, or their combination for panic disorder. JAMA 283, 2529. doi:10.1001/jama.283.19.2529.

Beck, A.T., Haigh, E.A., 2014. Advances in cognitive theory and therapy: The generic cognitive model. Annual Review of Clinical Psychology 10, 1–24. doi:10.1146/annurev-clinpsy-032813-153734.

Becker, C.B., Zayfert, C., Anderson, E., 2004. A survey of psychologists' attitudes towards and utilization of exposure therapy for ptsd. Behaviour Research and Therapy 42, 277–292. doi:10.1016/s0005-7967(03)00138-4.

Beidas, R.S., Kendall, P.C., 2010. Training therapists in evidence-based practice: A critical review of studies from a systems-contextual perspective. Clinical Psychology: Science and Practice 17, 1–30. doi:10.1111/j.1468-2850.2009.01187.x.

Benish, S.G., Quintana, S., Wampold, B.E., 2011. Culturally adapted psychotherapy and the legitimacy of myth: A direct-comparison meta-analysis. Journal of Counseling Psychology 58, 279–289. doi:10.1037/a0023626.

Berridge, K.C., Robinson, T.E., 2016. Liking, wanting, and the incentive-sensitization theory of addiction. American Psychologist 71, 670–679. doi:10.1037/amp0000059.

Beutler, L.E., Edwards, C., Someah, K., 2018. Adapting psychotherapy to patient reactance level: A meta-analytic review. Journal of Clinical Psychology 74, 1952–1963. doi:10.1002/jclp.22682.

Biesheuvel-Leliefeld, K.E., Kok, G.D., Bockting, C.L., Cuijpers, P., Hollon, S.D., van Marwijk, H.W., Smit, F., 2015. Effectiveness of psychological interventions in preventing recurrence of depressive disorder: Meta-analysis and meta-regression. Journal of Affective Disorders 174, 400–410. doi:10.1016/j.jad.2014.12.016.

Boesen, K., Paludan-Müller, A.S., Gøtzsche, P.C., Jørgensen, K.J., 2022. Extended-release methylphenidate for attention deficit hyperactivity disorder (adhd) in adults. Cochrane Database of Systematic Reviews 2022. doi:10.1002/14651858.cd012857.pub2.

Branson, A., Shafran, R., Myles, P., 2015. Investigating the relationship between competence and patient outcome with cbt. Behaviour Research and Therapy 68, 19–26. doi:10.1016/j.brat.2015.03.002.

Brembs, B., Lorenzetti, F.D., Reyes, F.D., Baxter, D.A., Byrne, J.H., 2002. Operant reward learning in Aplysia: Neuronal correlates and mechanisms. Science 296, 1706–1709. doi:10.1126/science.1069434.

Brewer, J., 2016. A simple way to break a bad habit. TEDMED talk. URL: https://drjud.com/a-simple-way-to-break-a-bad-habit/. transcript on the author's site. Popular talk; cited for the author's own claims.

Brewer, J., 2021. Unwinding Anxiety: New Science Shows How to Break the Cycles of Worry and Fear to Heal Your Mind. Avery. Popular-science book by a clinician-researcher. Cited for the author's own claims, not as primary evidence.

Brewer, J., 2026. Why CBT doesn't work for everyone with anxiety (and what to try next). drjud.com, dated 2026-02-10. URL: https://drjud.com/anxiety/cbt-not-working/. author's website; opinion, not peer reviewed.

Brewer, J.A., Roy, A., 2021. Can approaching anxiety like a habit lead to novel treatments? American Journal of Lifestyle Medicine 15, 489–494. doi:10.1177/15598276211008144.

Bryant, R.A., Moulds, M.L., Guthrie, R.M., Dang, S.T., Mastrodomenico, J., Nixon, R.D.V., Felmingham, K.L., Hopwood, S., Creamer, M., 2008. A randomized controlled trial of exposure therapy and cognitive restructuring for posttraumatic stress disorder. Journal of Consulting and Clinical Psychology 76, 695–703. doi:10.1037/a0012616.

Burlingame, G.M., Seebeck, J.D., Janis, R.A., Whitcomb, K.E., Barkowski, S., Rosendahl, J., Strauss, B., 2016. Outcome differences between individual and group formats when identical and nonidentical treatments, patients, and doses are compared: A 25-year meta-analytic perspective. Psychotherapy 53, 446–461. doi:10.1037/pst0000090.

Carlbring, P., Andersson, G., Cuijpers, P., Riper, H., Hedman-Lagerlöf, E., 2018. Internet-based vs. face-to-face cognitive behavior therapy for psychiatric and somatic disorders: an updated systematic review and meta-analysis. Cognitive Behaviour Therapy 47, 1–18. doi:10.1080/16506073.2017.1401115.

Carpenter, J.K., Andrews, L.A., Witcraft, S.M., Powers, M.B., Smits, J.A.J., Hofmann, S.G., 2018. Cognitive behavioral therapy for anxiety and related disorders: A meta-analysis of randomized placebo-controlled trials. Depression and Anxiety 35, 502–514. doi:10.1002/da.22728.

Cherkasova, M.V., French, L.R., Syer, C.A., Cousins, L., Galina, H., Ahmadi-Kashani, Y., Hechtman, L., 2020. Efficacy of cognitive behavioral therapy with and without medication for adults with adhd: A randomized clinical trial. Journal of Attention Disorders 24, 889–903. doi:10.1177/1087054716671197.

Chibanda, D., Weiss, H.A., Verhey, R., Simms, V., Munjoma, R., Rusakaniko, S., Chingono, A., Munetsi, E., Bere, T., Manda, E., Abas, M., Araya, R., 2016. Effect of a primary care–based psychological intervention on symptoms of common mental disorders in zimbabwe. JAMA 316, 2618. doi:10.1001/jama.2016.19102.

Ciharova, M., Furukawa, T.A., Efthimiou, O., Karyotaki, E., Miguel, C., Noma, H., Cipriani, A., Riper, H., Cuijpers, P., 2021. Cognitive restructuring, behavioral activation and cognitive-behavioral therapy in the treatment of adult depression: A network meta-analysis. Journal of Consulting and Clinical Psychology 89, 563–574. doi:10.1037/ccp0000654.

Clark, D.M., Canvin, L., Green, J., Layard, R., Pilling, S., Janecka, M., 2018. Transparency about the outcomes of mental health services (iapt approach): an analysis of public data. The Lancet 391, 679–686. doi:10.1016/s0140-6736(17)32133-5.

Constantino, M.J., Vîslă, A., Coyne, A.E., Boswell, J.F., 2018. A meta-analysis of the association between patients' early treatment outcome expectation and their posttreatment outcomes. Psychotherapy 55, 473–485. doi:10.1037/pst0000169.

Cooper, A.A., Conklin, L.R., 2015. Dropout from individual psychotherapy for major depression: A meta-analysis of randomized clinical trials. Clinical Psychology Review 40, 57–65. doi:10.1016/j.cpr.2015.05.001.

Cortese, S., Adamo, N., Del Giovane, C., Mohr-Jensen, C., Hayes, A.J., Carucci, S., Atkinson, L.Z., Tessari, L., Banaschewski, T., Coghill, D., Hollis, C., Simonoff, E., Zuddas, A., Barbui, C., Purgato, M., Steinhausen, H.C., Shokraneh, F., Xia, J., Cipriani, A., 2018. Comparative efficacy and tolerability of medications for attention-deficit hyperactivity disorder in children, adolescents, and adults: a systematic review and network meta-analysis. The Lancet Psychiatry 5, 727–738. doi:10.1016/S2215-0366(18)30269-4.

Craske, M.G., Treanor, M., Conway, C.C., Zbozinek, T., Vervliet, B., 2014. Maximizing exposure therapy: An inhibitory learning approach. Behaviour Research and Therapy 58, 10–23. doi:10.1016/j.brat.2014.04.006.

Crawford, M.J., Thana, L., Farquharson, L., Palmer, L., Hancock, E., Bassett, P., Clarke, J., Parry, G.D., 2016. Patient experience of negative effects of psychological treatment: results of a national survey. British Journal of Psychiatry 208, 260–265. doi:10.1192/bjp.bp.114.162628.

Cristea, I.A., Huibers, M.J., David, D., Hollon, S.D., Andersson, G., Cuijpers, P., 2015. The effects of cognitive behavior therapy for adult depression on dysfunctional thinking: A meta-analysis. Clinical Psychology Review 42, 62–71. doi:10.1016/j.cpr.2015.08.003.

Cristea, I.A., Stefan, S., Karyotaki, E., David, D., Hollon, S.D., Cuijpers, P., 2017. The effects of cognitive behavioral therapy are not systematically

falling: A revision of johnsen and friborg (2015). Psychological Bulletin 143, 326–340. doi:10.1037/bul0000062.

Cuijpers, P., 2015. The future of psychotherapy research: stop the waste and focus on issues that matter. Epidemiology and Psychiatric Sciences 25, 291–294. doi:10.1017/s2045796015000785.

Cuijpers, P., 2019. Targets and outcomes of psychotherapies for mental disorders: an overview. World Psychiatry 18, 276–285. doi:10.1002/wps.20661.

Cuijpers, P., Cristea, I.A., Karyotaki, E., Reijnders, M., Hollon, S.D., 2019a. Component studies of psychological treatments of adult depression: A systematic review and meta-analysis. Psychotherapy Research 29, 15–29. doi:10.1080/10503307.2017.1395922.

Cuijpers, P., Cristea, I.A., Karyotaki, E., Reijnders, M., Huibers, M.J., 2016. How effective are cognitive behavior therapies for major depression and anxiety disorders? a meta-analytic update of the evidence. World Psychiatry 15, 245–258. doi:10.1002/wps.20346.

Cuijpers, P., Harrer, M., Miguel, C., Ciharova, M., Papola, D., Basic, D., Botella, C., Cristea, I., de Ponti, N., Donker, T., et al., 2025. Cognitive behavior therapy for mental disorders in adults. JAMA Psychiatry 82, 563. doi:10.1001/jamapsychiatry.2025.0482.

Cuijpers, P., Hollon, S.D., van Straten, A., Bockting, C., Berking, M., Andersson, G., 2013a. Does cognitive behaviour therapy have an enduring effect that is superior to keeping patients on continuation pharmacotherapy? a meta-analysis. BMJ Open 3, e002542. doi:10.1136/bmjopen-2012-002542.

Cuijpers, P., Huibers, M., Daniel Ebert, D., Koole, S.L., Andersson, G., 2013b. How much psychotherapy is needed to treat depression? a metaregression analysis. Journal of Affective Disorders 149, 1–13. doi:10.1016/j.jad.2013.02.030.

Cuijpers, P., Karyotaki, E., Reijnders, M., Ebert, D.D., 2019b. Was eysenck right after all? a reassessment of the effects of psychotherapy for adult depression. Epidemiology and Psychiatric Sciences 28, 21–30. doi:10.1017/s2045796018000057.

Cuijpers, P., Karyotaki, E., Weitz, E., Andersson, G., Hollon, S.D., van Straten, A., 2014. The effects of psychotherapies for major depression in adults on remission, recovery and improvement: A meta-analysis. Journal of Affective Disorders 159, 118–126. doi:10.1016/j.jad.2014.02.026.

Cuijpers, P., Li, J., Hofmann, S.G., Andersson, G., 2010a. Self-reported versus clinician-rated symptoms of depression as outcome measures in psychotherapy research on depression: A meta-analysis. Clinical Psychology Review 30, 768–778. doi:10.1016/j.cpr.2010.06.001.

Cuijpers, P., Miguel, C., Harrer, M., Plessen, C.Y., Ciharova, M., Ebert, D., Karyotaki, E., 2023. Cognitive behavior therapy vs. control conditions, other psychotherapies, pharmacotherapies and combined treatment for depression: a comprehensive meta-analysis including 409 trials with 52,702 patients. World Psychiatry 22, 105–115. doi:10.1002/wps.21069.

Cuijpers, P., Reijnders, M., Huibers, M.J., 2019c. The role of common factors in psychotherapy outcomes. Annual Review of Clinical Psychology 15, 207–231. doi:10.1146/annurev-clinpsy-050718-095424.

Cuijpers, P., van Straten, A., Bohlmeijer, E., Hollon, S.D., Andersson, G., 2009. The effects of psychotherapy for adult depression are overestimated: a meta-analysis of study quality and effect size. Psychological Medicine 40, 211–223. doi:10.1017/s0033291709006114.

Cuijpers, P., van Straten, A., Schuurmans, J., van Oppen, P., Hollon, S.D., Andersson, G., 2010b. Psychotherapy for chronic major depression and dysthymia: A meta-analysis. Clinical Psychology Review 30, 51–62. doi:10.1016/j.cpr.2009.09.003.

Cybulski, L., Mayo-Wilson, E., Grant, S., 2016. Improving transparency and reproducibility through registration: The status of intervention trials published in clinical psychology journals. Journal of Consulting and Clinical Psychology 84, 753–767. doi:10.1037/ccp0000115.

Delgadillo, J., Ali, S., Fleck, K., Agnew, C., Southgate, A., Parkhouse, L., Cohen, Z.D., DeRubeis, R.J., Barkham, M., 2022. Stratified care vs stepped care for depression. JAMA Psychiatry 79, 101. doi:10.1001/jamapsychiatry.2021.3539.

Delgadillo, J., Asaria, M., Ali, S., Gilbody, S., 2016. On poverty, politics and psychology: the socioeconomic gradient of mental healthcare utilisation and outcomes. British Journal of Psychiatry 209, 429–430. doi:10.1192/bjp.bp.115.171017.

Delgadillo, J., de Jong, K., Lucock, M., Lutz, W., Rubel, J., Gilbody, S., Ali, S., Aguirre, E., Appleton, M., Nevin, J., O'Hayon, H., Patel, U., Sainty, A., Spencer, P., McMillan, D., 2018. Feedback-informed treatment versus usual psychological treatment for depression and anxiety: a multisite, open-label, cluster randomised controlled trial. The Lancet Psychiatry 5, 564–572. doi:10.1016/s2215-0366(18)30162-7.

Denny, B.T., Ochsner, K.N., 2014. Behavioral effects of longitudinal training in cognitive reappraisal. Emotion 14, 425–433. doi:10.1037/a0035276.

DeRubeis, R.J., Hollon, S.D., Amsterdam, J.D., Shelton, R.C., Young, P.R., Salomon, R.M., O'Reardon, J.P., Lovett, M.L., Gladis, M.M., Brown, L.L., Gallop, R., 2005. Cognitive therapy vs medications in the treatment of moderate to severe depression. Archives of General Psychiatry 62, 409. doi:10.1001/archpsyc.62.4.409.

van Dis, E.A.M., van Veen, S.C., Hagenaars, M.A., Batelaan, N.M., Bockting, C.L.H., van den Heuvel, R.M., Cuijpers, P., Engelhard, I.M., 2020. Long-term outcomes of cognitive behavioral therapy for anxiety-related disorders. JAMA Psychiatry 77, 265. doi:10.1001/jamapsychiatry.2019.3986.

Dolan, R.J., Dayan, P., 2013. Goals and habits in the brain. Neuron 80, 312–325. doi:10.1016/j.neuron.2013.09.007.

Doll, B.B., Jacobs, W.J., Sanfey, A.G., Frank, M.J., 2009. Instructional control of reinforcement learning: A behavioral and neurocomputational investigation. Brain Research 1299, 74–94. doi:10.1016/j.brainres.2009.07.007.

Driessen, E., Hollon, S.D., 2010. Cognitive behavioral therapy for mood disorders: Efficacy, moderators and mediators. Psychiatric Clinics of North America 33, 537–555. doi:10.1016/j.psc.2010.04.005.

Driessen, E., Hollon, S.D., Bockting, C.L.H., Cuijpers, P., Turner, E.H., 2015. Does publication bias inflate the apparent efficacy of psychological

treatment for major depressive disorder? a systematic review and meta-analysis of us national institutes of health-funded trials. PLOS ONE 10, e0137864. doi:10.1371/journal.pone.0137864.

Duggan, C., Parry, G., McMurran, M., Davidson, K., Dennis, J., 2014. The recording of adverse events from psychological treatments in clinical trials: evidence from a review of nihr-funded trials. Trials 15, 335. doi:10.1186/1745-6215-15-335.

Ebert, D.D., Donkin, L., Andersson, G., Andrews, G., Berger, T., Carlbring, P., et al., 2016. Does internet-based guided-self-help for depression cause harm? an individual participant data meta-analysis on deterioration rates and its moderators in randomized controlled trials. Psychological Medicine 46, 2679–2693. doi:10.1017/s0033291716001562.

Emerson, G.B., Warme, W.J., Wolf, F.M., Heckman, J.D., Brand, R.A., Leopold, S.S., 2010. Testing for the presence of positive-outcome bias in peer review. Archives of Internal Medicine 170, 1934–1939. doi:10.1001/archinternmed.2010.406.

Enck, P., Zipfel, S., 2019. Placebo effects in psychotherapy: A framework. Frontiers in Psychiatry 10, 456. doi:10.3389/fpsyt.2019.00456.

Etzelmueller, A., Vis, C., Karyotaki, E., Baumeister, H., Titov, N., Berking, M., Cuijpers, P., Riper, H., Ebert, D.D., 2020. Effects of internet-based cognitive behavioral therapy in routine care for adults in treatment for depression and anxiety: Systematic review and meta-analysis. Journal of Medical Internet Research 22, e18100. doi:10.2196/18100.

Eubanks, C.F., Muran, J.C., Safran, J.D., 2018. Alliance rupture repair: A meta-analysis. Psychotherapy 55, 508–519. doi:10.1037/pst0000185.

Eysenck, M.W., Derakshan, N., Santos, R., Calvo, M.G., 2007. Anxiety and cognitive performance: Attentional control theory. Emotion 7, 336–353. doi:10.1037/1528-3542.7.2.336.

Ezawa, I.D., Hollon, S.D., 2023. Cognitive restructuring and psychotherapy outcome: A meta-analytic review. Psychotherapy 60, 396–406. doi:10.1037/pst0000474.

Fernandez, E., Salem, D., Swift, J.K., Ramtahal, N., 2015. Meta-analysis of dropout from cognitive behavioral therapy: Magnitude, timing, and moderators. Journal of Consulting and Clinical Psychology 83, 1108–1122. doi:10.1037/ccp0000044.

Firth, N., Barkham, M., Kellett, S., 2015a. The clinical effectiveness of stepped care systems for depression in working age adults: A systematic review. Journal of Affective Disorders 170, 119–130. doi:10.1016/j.jad.2014.08.030.

Firth, N., Barkham, M., Kellett, S., Saxon, D., 2015b. Therapist effects and moderators of effectiveness and efficiency in psychological wellbeing practitioners: A multilevel modelling analysis. Behaviour Research and Therapy 69, 54–62. doi:10.1016/j.brat.2015.04.001.

Flint, J., Cuijpers, P., Horder, J., Koole, S.L., Munafò, M.R., 2014. Is there an excess of significant findings in published studies of psychotherapy for depression? Psychological Medicine 45, 439–446. doi:10.1017/s0033291714001421.

Flückiger, C., Del Re, A.C., Wampold, B.E., Horvath, A.O., 2018. The alliance in adult psychotherapy: A meta-analytic synthesis. Psychotherapy 55, 316–340. doi:10.1037/pst0000172.

Flückiger, C., Rubel, J., Del Re, A.C., Horvath, A.O., Wampold, B.E., Crits-Christoph, P., Atzil-Slonim, D., Compare, A., Falkenström, F., Ekeblad, A., Errázuriz, P., Fisher, H., Hoffart, A., Huppert, J.D., Kivity, Y., Kumar, M., Lutz, W., Muran, J.C., Strunk, D.R., Tasca, G.A., Vîslă, A., Voderholzer, U., Webb, C.A., Xu, H., Zilcha-Mano, S., Barber, J.P., 2020. The reciprocal relationship between alliance and early treatment symptoms: A two-stage individual participant data meta-analysis. Journal of Consulting and Clinical Psychology 88, 829–843. doi:10.1037/ccp0000594.

Foa, E.B., Hembree, E.A., Cahill, S.P., Rauch, S.A.M., Riggs, D.S., Feeny, N.C., Yadin, E., 2005. Randomized trial of prolonged exposure for posttraumatic stress disorder with and without cognitive restructuring: Outcome at academic and community clinics. Journal of Consulting and Clinical Psychology 73, 953–964. doi:10.1037/0022-006x.73.5.953.

Fried, E.I., 2017. The 52 symptoms of major depression: Lack of content overlap among seven common depression scales. Journal of Affective Disorders 208, 191–197. doi:10.1016/j.jad.2016.10.019.

Fried, E.I., van Borkulo, C.D., Epskamp, S., Schoevers, R.A., Tuerlinckx, F., Borsboom, D., 2016. Measuring depression over time . . . or not? lack of unidimensionality and longitudinal measurement invariance in four common rating scales of depression. Psychological Assessment 28, 1354–1367. doi:10.1037/pas0000275.

Fried, E.I., Nesse, R.M., 2015. Depression sum-scores don't add up: why analyzing specific depression symptoms is essential. BMC Medicine 13, 72. doi:10.1186/s12916-015-0325-4.

Gan, S., Yang, J., Chen, X., Zhang, X., Yang, Y., 2017. High working memory load impairs the effect of cognitive reappraisal on emotional response: Evidence from an event-related potential study. Neuroscience Letters 639, 126–131. doi:10.1016/j.neulet.2016.12.069.

Gilbody, S., Littlewood, E., Hewitt, C., Brierley, G., Tharmanathan, P., Araya, R., et al., 2015. Computerised cognitive behaviour therapy (ccbt) as treatment for depression in primary care (reeact trial): large scale pragmatic randomised controlled trial. BMJ , h5627doi:10.1136/bmj.h5627.

van Ginneken, N., Chin, W.Y., Lim, Y.C., Ussif, A., Singh, R., Shahmalak, U., Purgato, M., Rojas-García, A., Uphoff, E., McMullen, S., Foss, H.S., Thapa Pachya, A., Rashidian, L., Borghesani, A., Henschke, N., Chong, L.Y., Lewin, S., 2021. Primary-level worker interventions for the care of people living with mental disorders and distress in low- and middle-income countries. Cochrane Database of Systematic Reviews 2021. doi:10.1002/14651858.cd009149.pub3.

Godovich, S.A., Senior, C.J., Degnan, K.A., Cummings, C., Shiffrin, N.D., Alvord, M.K., Rich, B.A., 2020. The role of executive functioning in treatment outcome for child anxiety. Evidence-Based Practice in Child and Adolescent Mental Health 5, 53–66. doi:10.1080/23794925.2020.1727794.

Goldberg, S.B., Rousmaniere, T., Miller, S.D., Whipple, J., Nielsen, S.L., Hoyt, W.T., Wampold, B.E., 2016. Do psychotherapists improve with time

and experience? a longitudinal analysis of outcomes in a clinical setting. Journal of Counseling Psychology 63, 1–11. doi:10.1037/cou0000131.

Goldberg, S.B., Tucker, R.P., Greene, P.A., Davidson, R.J., Wampold, B.E., Kearney, D.J., Simpson, T.L., 2018. Mindfulness-based interventions for psychiatric disorders: A systematic review and meta-analysis. Clinical Psychology Review 59, 52–60. doi:10.1016/j.cpr.2017.10.011.

Goodkind, M.S., Gallagher-Thompson, D., Thompson, L.W., Kesler, S.R., Anker, L., Flournoy, J., Berman, M.P., Holland, J.M., O'Hara, R.M., 2016. The impact of executive function on response to cognitive behavioral therapy in late-life depression. International Journal of Geriatric Psychiatry 31, 334–339. doi:10.1002/gps.4325.

Gosling, C.J., Garcia-Argibay, M., De Prisco, M., Arrondo, G., Ayrolles, A., Antoun, S., Caparos, S., Catalán, A., Ellul, P., Dobrosavljevic, M., et al., 2025. Benefits and harms of adhd interventions: umbrella review and platform for shared decision making. BMJ 391, e085875. doi:10.1136/bmj-2025-085875.

Gould, K.L., Porter, M., Lyneham, H.J., Hudson, J.L., 2018. Cognitive-behavioral therapy for children with anxiety and comorbid attention-deficit/hyperactivity disorder. Journal of the American Academy of Child & Adolescent Psychiatry 57, 481–490.e2. doi:10.1016/j.jaac.2018.03.021.

Grillner, S., Robertson, B., 2016. The basal ganglia over 500 million years. Current Biology 26, R1088–R1100. doi:10.1016/j.cub.2016.06.041.

Gyani, A., Shafran, R., Layard, R., Clark, D.M., 2013. Enhancing recovery rates: Lessons from year one of iapt. Behaviour Research and Therapy 51, 597–606. doi:10.1016/j.brat.2013.06.004.

Hall, G.C.N., Ibaraki, A.Y., Huang, E.R., Marti, C.N., Stice, E., 2016. A meta-analysis of cultural adaptations of psychological interventions. Behavior Therapy 47, 993–1014. doi:10.1016/j.beth.2016.09.005.

Hallion, L.S., Tolin, D.F., Diefenbach, G.J., 2019. Enhanced cognitive control over task-irrelevant emotional distractors in generalized anxiety disorder versus obsessive-compulsive disorder. Journal of Anxiety Disorders 64, 71–78. doi:10.1016/j.janxdis.2019.02.004.

Hamza, J., Vytykačová, S., Janšáková, K., Rajčáni, J., 2024. Cognitive reappraisal and acceptance following acute stress. Stress and Health 40, e3469. doi:10.1002/smi.3469.

Hans, E., Hiller, W., 2013. Effectiveness of and dropout from outpatient cognitive behavioral therapy for adult unipolar depression: A meta-analysis of nonrandomized effectiveness studies. Journal of Consulting and Clinical Psychology 81, 75–88. doi:10.1037/a0031080.

Harrison, A.A., Nguyen, L., Murphy, K., Neumann, D., 2025. Assessing executive functioning in individuals with social anxiety disorder (sad) across the lifespan: A systematic literature review and meta-analysis. Journal of Anxiety Disorders 115, 103056. doi:10.1016/j.janxdis.2025.103056.

Hayes, S.C., Hofmann, S.G., 2021. "third-wave" cognitive and behavioral therapies and the emergence of a process-based approach to intervention in psychiatry. World Psychiatry 20, 363–375. doi:10.1002/wps.20884.

Hayes, S.C., Luoma, J.B., Bond, F.W., Masuda, A., Lillis, J., 2006. Acceptance and commitment therapy: Model, processes and outcomes. Behaviour Research and Therapy 44, 1–25. doi:10.1016/j.brat.2005.06.006.

Hayes, S.C., Wilson, K.G., Gifford, E.V., Follette, V.M., Strosahl, K., 1996. Experiential avoidance and behavioral disorders: A functional dimensional approach to diagnosis and treatment. Journal of Consulting and Clinical Psychology 64, 1152–1168. doi:10.1037/0022-006x.64.6.1152.

Henrich, J., Heine, S.J., Norenzayan, A., 2010. The weirdest people in the world? Behavioral and Brain Sciences 33, 61–83. doi:10.1017/s0140525x0999152x.

Hermans, E.J., Henckens, M.J., Joëls, M., Fernández, G., 2014. Dynamic adaptation of large-scale brain networks in response to acute stressors. Trends in Neurosciences 37, 304–314. doi:10.1016/j.tins.2014.03.006.

Hofmann, S.G., 2008. Cognitive processes during fear acquisition and extinction in animals and humans: Implications for exposure therapy of anxiety disorders. Clinical Psychology Review 28, 199–210. doi:10.1016/j.cpr.2007.04.009.

Hofmann, S.G., Asmundson, G.J., Beck, A.T., 2013. The science of cognitive therapy. Behavior Therapy 44, 199–212. doi:10.1016/j.beth.2009.01.007.

Hofmann, S.G., Hayes, S.C., 2018. The future of intervention science: Process-based therapy. Clinical Psychological Science 7, 37–50. doi:10.1177/2167702618772296.

Hofmann, S.G., Heering, S., Sawyer, A.T., Asnaani, A., 2009. How to handle anxiety: The effects of reappraisal, acceptance, and suppression strategies on anxious arousal. Behaviour Research and Therapy 47, 389–394. doi:10.1016/j.brat.2009.02.010.

Hofmann, S.G., Kasch, C., Reis, A., 2025. Effect sizes of randomized-controlled studies of cognitive behavioral therapy for anxiety disorders over the past 30 years. Clinical Psychology Review 117, 102553. doi:10.1016/j.cpr.2025.102553.

Hoge, E.A., Bui, E., Mete, M., Dutton, M.A., Baker, A.W., Simon, N.M., 2023. Mindfulness-based stress reduction vs escitalopram for the treatment of adults with anxiety disorders. JAMA Psychiatry 80, 13. doi:10.1001/jamapsychiatry.2022.3679.

Hollon, S.D., DeRubeis, R.J., Shelton, R.C., Amsterdam, J.D., Salomon, R.M., O'Reardon, J.P., Lovett, M.L., Young, P.R., Haman, K.L., Freeman, B.B., Gallop, R., 2005. Prevention of relapse following cognitive therapy vs medications in moderate to severe depression. Archives of General Psychiatry 62, 417. doi:10.1001/archpsyc.62.4.417.

Hoskins, J.I., Gooch, D., Gilligan-Lee, K.A., 2026. Symptoms of anxiety in adults with adhd: A systematic review and meta-analysis of case-control studies. Journal of Attention Disorders 30, 1305–1337. doi:10.1177/10870547261459712.

Hundt, N.E., Mignogna, J., Underhill, C., Cully, J.A., 2013. The relationship between use of cbt skills and depression treatment outcome: A theoretical and methodological review of the literature. Behavior Therapy 44, 12–26. doi:10.1016/j.beth.2012.10.001.

Hybel, K.A., Mortensen, E.L., Lambek, R., Højgaard, D.R., Thomsen, P.H., 2017. Executive function predicts cognitive-behavioral therapy response in

childhood obsessive-compulsive disorder. Behaviour Research and Therapy 99, 11–18. doi:10.1016/j.brat.2017.08.009.

Ijaz, S., Davies, P., Williams, C.J., Kessler, D., Lewis, G., Wiles, N., 2018. Psychological therapies for treatment-resistant depression in adults. Cochrane Database of Systematic Reviews 2018. doi:10.1002/14651858.cd010558.pub2.

Jacobson, N.S., Dobson, K.S., Truax, P.A., Addis, M.E., Koerner, K., Gollan, J.K., Gortner, E., Prince, S.E., 1996. A component analysis of cognitive-behavioral treatment for depression. Journal of Consulting and Clinical Psychology 64, 295–304. doi:10.1037/0022-006X.64.2.295.

Jakobsen, J.C., Lindschou Hansen, J., Storebø, O.J., Simonsen, E., Gluud, C., 2011. The effects of cognitive therapy versus 'treatment as usual' in patients with major depressive disorder. PLoS ONE 6, e22890. doi:10.1371/journal.pone.0022890.

JAMA Psychiatry, 2024. Errors in network meta-analysis. JAMA Psychiatry 81, 320. doi:10.1001/jamapsychiatry.2023.5480.

Jarrett, R.B., Kraft, D., Doyle, J., Foster, B.M., Eaves, G.G., Silver, P.C., 2001. Preventing recurrent depression using cognitive therapy with and without a continuation phase. Archives of General Psychiatry 58, 381. doi:10.1001/archpsyc.58.4.381.

Johnco, C., Wuthrich, V., Rapee, R., 2014. The influence of cognitive flexibility on treatment outcome and cognitive restructuring skill acquisition during cognitive behavioural treatment for anxiety and depression in older adults: Results of a pilot study. Behaviour Research and Therapy 57, 55–64. doi:10.1016/j.brat.2014.04.005.

Johnco, C.J., Dickson, S.J., Seaton, A., 2026. A systematic review and meta-analysis of diagnostic remission, treatment response, attrition and relapse following cognitive behavior therapy (cbt), other psychological therapies and pharmacological treatments for anxiety disorders in older adults. The American Journal of Geriatric Psychiatry 34, 500–521. doi:10.1016/j.jagp.2025.05.010.

Johnco, C.J., Norberg, M., Wuthrich, V.M., Rapee, R.M., 2025. Cognitive restructuring before exposure therapy or behavioral experiments? how

the timing of expectancy violation and magnitude of expectancy change influence exposure therapy outcomes. Journal of Consulting and Clinical Psychology 93, 369–381. doi:10.1037/ccp0000953.

Johns, R.G., Barkham, M., Kellett, S., Saxon, D., 2019. A systematic review of therapist effects: A critical narrative update and refinement to review. Clinical Psychology Review 67, 78–93. doi:10.1016/j.cpr.2018.08.004.

Johnsen, T.J., Friborg, O., 2015. The effects of cognitive behavioral therapy as an anti-depressive treatment is falling: A meta-analysis. Psychological Bulletin 141, 747–768. doi:10.1037/bul0000015.

Jonsson, U., Alaie, I., Parling, T., Arnberg, F.K., 2014. Reporting of harms in randomized controlled trials of psychological interventions for mental and behavioral disorders: A review of current practice. Contemporary Clinical Trials 38, 1–8. doi:10.1016/j.cct.2014.02.005.

Karyotaki, E., Efthimiou, O., Miguel, C., Maas genannt Bermpohl, F., Furukawa, T.A., Cuijpers, P., et al., 2021. Internet-based cognitive behavioral therapy for depression. JAMA Psychiatry 78, 361. doi:10.1001/jamapsychiatry.2020.4364.

Karyotaki, E., Kleiboer, A., Smit, F., Turner, D.T., Pastor, A.M., Andersson, G., et al., 2015. Predictors of treatment dropout in self-guided web-based interventions for depression: an 'individual patient data' meta-analysis. Psychological Medicine 45, 2717–2726. doi:10.1017/s0033291715000665.

Karyotaki, E., Riper, H., Twisk, J., Hoogendoorn, A., Kleiboer, A., Mira, A., et al., 2017. Efficacy of self-guided internet-based cognitive behavioral therapy in the treatment of depressive symptoms. JAMA Psychiatry 74, 351. doi:10.1001/jamapsychiatry.2017.0044.

Kazantzis, N., Whittington, C., Zelencich, L., Kyrios, M., Norton, P.J., Hofmann, S.G., 2016. Quantity and quality of homework compliance: A meta-analysis of relations with outcome in cognitive behavior therapy. Behavior Therapy 47, 755–772. doi:10.1016/j.beth.2016.05.002.

Kazdin, A.E., 2007. Mediators and mechanisms of change in psychotherapy research. Annual Review of Clinical Psychology 3, 1–27. doi:10.1146/annurev.clinpsy.3.022806.091432.

Kazdin, A.E., 2009. Understanding how and why psychotherapy leads to change. Psychotherapy Research 19, 418–428. doi:10.1080/10503300802448899.

Khoury, B., Lecomte, T., Fortin, G., Masse, M., Therien, P., Bouchard, V., Chapleau, M.A., Paquin, K., Hofmann, S.G., 2013. Mindfulness-based therapy: A comprehensive meta-analysis. Clinical Psychology Review 33, 763–771. doi:10.1016/j.cpr.2013.05.005.

Klatte, R., Strauss, B., Flückiger, C., Rosendahl, J., 2023. Adverse events in psychotherapy randomized controlled trials: A systematic review. Psychotherapy Research 35, 84–99. doi:10.1080/10503307.2023.2286992.

Knouse, L.E., Safren, S.A., 2010. Current status of cognitive behavioral therapy for adult attention-deficit hyperactivity disorder. Psychiatric Clinics of North America 33, 497–509. doi:10.1016/j.psc.2010.04.001.

Knouse, L.E., Teller, J., Brooks, M.A., 2017. Meta-analysis of cognitive–behavioral treatments for adult adhd. Journal of Consulting and Clinical Psychology 85, 737–750. doi:10.1037/ccp0000216.

Kofler, M.J., Irwin, L.N., Soto, E.F., Groves, N.B., Harmon, S.L., Sarver, D.E., 2019. Executive functioning heterogeneity in pediatric adhd. Journal of Abnormal Child Psychology 47, 273–286. doi:10.1007/s10802-018-0438-2.

Kooij, J., Bijlenga, D., Salerno, L., Jaeschke, R., Bitter, I., Balázs, J., Thome, J., Dom, G., Kasper, S., Nunes Filipe, C., et al., 2019. Updated european consensus statement on diagnosis and treatment of adult adhd. European Psychiatry 56, 14–34. doi:10.1016/j.eurpsy.2018.11.001.

Kraemer, H.C., Wilson, G.T., Fairburn, C.G., Agras, W.S., 2002. Mediators and moderators of treatment effects in randomized clinical trials. Archives of General Psychiatry 59, 877. doi:10.1001/archpsyc.59.10.877.

Krause, K.L., Koerner, N., Antony, M.M., 2022. Cognitive restructuring before versus after exposure: Effect on expectancy and outcome in individuals with claustrophobia. Behavior Modification 46, 1432–1459. doi:10.1177/01454455221075754.

Krebs, P., Norcross, J.C., Nicholson, J.M., Prochaska, J.O., 2018. Stages of change and psychotherapy outcomes: A review and meta-analysis. Journal of Clinical Psychology 74, 1964–1979. doi:10.1002/jclp.22683.

Kuyken, W., Tsivrikos, D., 2008. Therapist competence, comorbidity and cognitive-behavioral therapy for depression. Psychotherapy and Psychosomatics 78, 42–48. doi:10.1159/000172619.

Kyriacou, T., Hodges, J., Gould, R.L., 2023. Predictors and moderators of treatment outcome in late-life anxiety: A systematic review. Journal of Affective Disorders 339, 454–470. doi:10.1016/j.jad.2023.07.057.

Lakdawalla, Z., Hankin, B.L., Mermelstein, R., 2007. Cognitive theories of depression in children and adolescents: A conceptual and quantitative review. Clinical Child and Family Psychology Review 10, 1–24. doi:10.1007/s10567-006-0013-1.

Lambert, M.J., Ogles, B.M., 2009. Using clinical significance in psychotherapy outcome research: The need for a common procedure and validity data. Psychotherapy Research 19, 493–501. doi:10.1080/10503300902849483.

Lambert, M.J., Whipple, J.L., Kleinstäuber, M., 2018. Collecting and delivering progress feedback: A meta-analysis of routine outcome monitoring. Psychotherapy 55, 520–537. doi:10.1037/pst0000167.

Langer, K., Jentsch, V.L., Wolf, O.T., 2023. Rapid effects of acute stress on cognitive emotion regulation. Psychoneuroendocrinology 151, 106054. doi:10.1016/j.psyneuen.2023.106054.

Langer, K., Wolf, O.T., Jentsch, V.L., 2021. Delayed effects of acute stress on cognitive emotion regulation. Psychoneuroendocrinology 125, 105101. doi:10.1016/j.psyneuen.2020.105101.

Leichsenring, F., Steinert, C., Rabung, S., Ioannidis, J.P., 2022. The efficacy of psychotherapies and pharmacotherapies for mental disorders in adults: an umbrella review and meta-analytic evaluation of recent meta-analyses. World Psychiatry 21, 133–145. doi:10.1002/wps.20941.

Li, J., Delgado, M.R., Phelps, E.A., 2011. How instructed knowledge modulates the neural systems of reward learning. Proceedings of the National Academy of Sciences 108, 55–60. doi:10.1073/pnas.1014938108.

Li, S., Xi, Z., Barnett, P., Saunders, R., Shafran, R., Pilling, S., 2023. Efficacy of culturally adapted interventions for common mental disorders in people of chinese descent: a systematic review and meta-analysis. The Lancet Psychiatry 10, 426–440. doi:10.1016/s2215-0366(23)00118-9.

Lilienfeld, S.O., 2007. Psychological treatments that cause harm. Perspectives on Psychological Science 2, 53–70. doi:10.1111/j.1745-6916.2007.00029.x.

Linden, M., Schermuly-Haupt, M.L., 2014. Definition, assessment and rate of psychotherapy side effects. World Psychiatry 13, 306–309. doi:10.1002/wps.20153.

Liu, Y., Zhu, F., Yu, Y., Song, Y., Luo, H., Xu, B., Lin, S., Ha, Z., Kuang, D., Xie, Y., Chen, Z., Ren, Y., 2026. A meta-analysis of the intervention effect of cognitive behavioral therapy on adult adhd. Journal of Affective Disorders 399, 121107. doi:10.1016/j.jad.2025.121107.

Loerinc, A.G., Meuret, A.E., Twohig, M.P., Rosenfield, D., Bluett, E.J., Craske, M.G., 2015. Response rates for cbt for anxiety disorders: Need for standardized criteria. Clinical Psychology Review 42, 72–82. doi:10.1016/j.cpr.2015.08.004.

Lopez, P.L., Torrente, F.M., Ciapponi, A., Lischinsky, A.G., Cetkovich-Bakmas, M., Rojas, J.I., Romano, M., Manes, F.F., 2018. Cognitive-behavioural interventions for attention deficit hyperactivity disorder (adhd) in adults. Cochrane Database of Systematic Reviews 2018. doi:10.1002/14651858.CD010840.pub2.

Lorenzo-Luaces, L., German, R.E., DeRubeis, R.J., 2015. It's complicated: The relation between cognitive change procedures, cognitive change, and symptom change in cognitive therapy for depression. Clinical Psychology Review 41, 3–15. doi:10.1016/j.cpr.2014.12.003.

Ludwig, V.U., Brown, K.W., Brewer, J.A., 2020. Self-regulation without force: Can awareness leverage reward to drive behavior change? Perspectives on Psychological Science 15, 1382–1399. doi:10.1177/1745691620931460.

López-Pinar, C., Martínez-Sanchís, S., Carbonell-Vayá, E., Sánchez-Meca, J., Fenollar-Cortés, J., 2020. Efficacy of nonpharmacological treat-

ments on comorbid internalizing symptoms of adults with attention-deficit/hyperactivity disorder: A meta-analytic review. Journal of Attention Disorders 24, 456–478. doi:10.1177/1087054719855685.

Magill, M., Ray, L., Kiluk, B., Hoadley, A., Bernstein, M., Tonigan, J.S., Carroll, K., 2019. A meta-analysis of cognitive-behavioral therapy for alcohol or other drug use disorders: Treatment efficacy by contrast condition. Journal of Consulting and Clinical Psychology 87, 1093–1105. doi:10.1037/ccp0000447.

Marsh, C.L., Harmon, S.L., Cho, S., Chan, E.S.M., Gaye, F., DeGeorge, L., Black, K.E., Irwin Harper, L.N., Kofler, M.J., 2024. Does anxiety systematically bias estimates of executive functioning deficits in pediatric attention-deficit/hyperactivity disorder? Research on Child and Adolescent Psychopathology 52, 773–787. doi:10.1007/s10802-023-01152-y.

Mausbach, B.T., Moore, R., Roesch, S., Cardenas, V., Patterson, T.L., 2010. The relationship between homework compliance and therapy outcomes: An updated meta-analysis. Cognitive Therapy and Research 34, 429–438. doi:10.1007/s10608-010-9297-z.

Mayo-Wilson, E., Dias, S., Mavranezouli, I., Kew, K., Clark, D.M., Ades, A.E., Pilling, S., 2014. Psychological and pharmacological interventions for social anxiety disorder in adults: a systematic review and network meta-analysis. The Lancet Psychiatry 1, 368–376. doi:10.1016/S2215-0366(14)70329-3.

Miguel, C., Karyotaki, E., Cuijpers, P., Cristea, I.A., 2021. Selective outcome reporting and the effectiveness of psychotherapies for depression. World Psychiatry 20, 444–445. doi:10.1002/wps.20900.

Miyake, A., Friedman, N.P., Emerson, M.J., Witzki, A.H., Howerter, A., Wager, T.D., 2000. The unity and diversity of executive functions and their contributions to complex "frontal lobe" tasks: A latent variable analysis. Cognitive Psychology 41, 49–100. doi:10.1006/cogp.1999.0734.

Mohlman, J., Gorman, J.M., 2005. The role of executive functioning in cbt: a pilot study with anxious older adults. Behaviour Research and Therapy 43, 447–465. doi:10.1016/j.brat.2004.03.007.

de Mooij, B., Fekkes, M., Miers, A.C., van den Akker, A.L., Scholte, R.H.J., Overbeek, G., 2023. What works in preventing emerging social anxiety:

Exposure, cognitive restructuring, or a combination? Journal of Child and Family Studies 32, 498–515. doi:10.1007/s10826-023-02536-w.

Moran, T.P., 2016. Anxiety and working memory capacity: A meta-analysis and narrative review. Psychological Bulletin 142, 831–864. doi:10.1037/bul0000051.

Munder, T., Barth, J., 2017. Cochrane's risk of bias tool in the context of psychotherapy outcome research. Psychotherapy Research 28, 347–355. doi:10.1080/10503307.2017.1411628.

Munder, T., Brütsch, O., Leonhart, R., Gerger, H., Barth, J., 2013. Researcher allegiance in psychotherapy outcome research: An overview of reviews. Clinical Psychology Review 33, 501–511. doi:10.1016/j.cpr.2013.02.002.

Nanni, V., Uher, R., Danese, A., 2012. Childhood maltreatment predicts unfavorable course of illness and treatment outcome in depression: A meta-analysis. American Journal of Psychiatry 169, 141–151. doi:10.1176/appi.ajp.2011.11020335.

National Institute for Health and Care Excellence, 2018. Attention deficit hyperactivity disorder: diagnosis and management (ng87). URL: https://www.nice.org.uk/guidance/ng87. published 14 March 2018; last updated 13 September 2019.

Newby, J.M., McKinnon, A., Kuyken, W., Gilbody, S., Dalgleish, T., 2015. Systematic review and meta-analysis of transdiagnostic psychological treatments for anxiety and depressive disorders in adulthood. Clinical Psychology Review 40, 91–110. doi:10.1016/j.cpr.2015.06.002.

Newton-Howes, G., Tyrer, P., Johnson, T., Mulder, R., Kool, S., Dekker, J., Schoevers, R., 2014. Influence of personality on the outcome of treatment in depression: Systematic review and meta-analysis. Journal of Personality Disorders 28, 577–593. doi:10.1521/pedi_2013_27_070.

Nguyen, L., Walters, J., Hutchinson, E., Liu, Y., Li, X., Gudmundsson, C., 2025. Executive functioning in individuals with generalized anxiety disorder: A systematic review and meta-analysis. Journal of Affective Disorders 389, 119683. doi:10.1016/j.jad.2025.119683.

Nolen-Hoeksema, S., Wisco, B.E., Lyubomirsky, S., 2008. Rethinking rumination. Perspectives on Psychological Science 3, 400–424. doi:10.1111/j.1745-6924.2008.00088.x.

Norton, P.J., Price, E.C., 2007. A meta-analytic review of adult cognitive-behavioral treatment outcome across the anxiety disorders. Journal of Nervous & Mental Disease 195, 521–531. doi:10.1097/01.nmd.0000253843.70149.9a.

Olatunji, B.O., Cisler, J.M., Tolin, D.F., 2010. A meta-analysis of the influence of comorbidity on treatment outcome in the anxiety disorders. Clinical Psychology Review 30, 642–654. doi:10.1016/j.cpr.2010.04.008.

Orgeta, V., Leung, P., del Pino-Casado, R., Qazi, A., Orrell, M., Spector, A.E., Methley, A.M., 2022. Psychological treatments for depression and anxiety in dementia and mild cognitive impairment. Cochrane Database of Systematic Reviews 2022. doi:10.1002/14651858.cd009125.pub3.

Ostinelli, E.G., Schulze, M., Zangani, C., Farhat, L.C., Tomlinson, A., Del Giovane, C., Chamberlain, S.R., Philipsen, A., Young, S., Cowen, P.J., Bilbow, A., Cipriani, A., Cortese, S., 2025. Comparative efficacy and acceptability of pharmacological, psychological, and neurostimulatory interventions for adhd in adults: a systematic review and component network meta-analysis. The Lancet Psychiatry 12, 32–43. doi:10.1016/s2215-0366(24)00360-2.

Ougrin, D., 2011. Efficacy of exposure versus cognitive therapy in anxiety disorders: systematic review and meta-analysis. BMC Psychiatry 11, 200. doi:10.1186/1471-244X-11-200.

Papola, D., Miguel, C., Mazzaglia, M., Franco, P., Tedeschi, F., Romero, S.A., Patel, A.R., Ostuzzi, G., Gastaldon, C., Karyotaki, E., Harrer, M., Purgato, M., Sijbrandij, M., Patel, V., Furukawa, T.A., Cuijpers, P., Barbui, C., 2024. Psychotherapies for generalized anxiety disorder in adults. JAMA Psychiatry 81, 250. doi:10.1001/jamapsychiatry.2023.3971.

Patel, V., Weobong, B., Weiss, H.A., Anand, A., Bhat, B., Katti, B., Dimidjian, S., Araya, R., Hollon, S.D., King, M., Vijayakumar, L., Park, A.L., McDaid, D., Wilson, T., Velleman, R., Kirkwood, B.R., Fairburn, C.G., 2017. The healthy activity program (hap), a lay counsellor-delivered brief psychological treatment for severe depression, in primary

care in india: a randomised controlled trial. The Lancet 389, 176–185. doi:10.1016/s0140-6736(16)31589-6.

Perez-Castilla, J., Contreras-Pizarro, C.H., Valdivieso-Jiménez, G., Munoz-Figueroa, A., Sifuentes Montalvo, A.V., Taype-Rondan, A., Brañez-Condorena, A., 2026. Effects of cognitive behavioral therapy in autistic adults: A systematic review and meta-analysis of randomized controlled trials. Autism in Adulthood , 25739581261450541doi:10.1177/25739581261450541.

Philipsen, A., Jans, T., Graf, E., Matthies, S., Borel, P., Colla, M., Gentschow, L., Langner, D., Jacob, C., Groß-Lesch, S., et al., 2015. Effects of group psychotherapy, individual counseling, methylphenidate, and placebo in the treatment of adult attention-deficit/hyperactivity disorder. JAMA Psychiatry 72, 1199. doi:10.1001/jamapsychiatry.2015.2146.

Pimontel, M.A., Rindskopf, D., Rutherford, B.R., Brown, P.J., Roose, S.P., Sneed, J.R., 2016. A meta-analysis of executive dysfunction and antidepressant treatment response in late-life depression. The American Journal of Geriatric Psychiatry 24, 31–41. doi:10.1016/j.jagp.2015.05.010.

Pittig, A., Heinig, I., Goerigk, S., Thiel, F., Hummel, K., Scholl, L., et al., 2021. Efficacy of temporally intensified exposure for anxiety disorders: A multicenter randomized clinical trial. Depression and Anxiety 38, 1169–1181. doi:10.1002/da.23204.

Plessen, C.Y., Karyotaki, E., Miguel, C., Ciharova, M., Cuijpers, P., 2023. Exploring the efficacy of psychotherapies for depression: a multiverse meta-analysis. BMJ Mental Health 26, e300626. doi:10.1136/bmjment-2022-300626.

Polo, A.J., Makol, B.A., Castro, A.S., Colón-Quintana, N., Wagstaff, A.E., Guo, S., 2019. Diversity in randomized clinical trials of depression: A 36-year review. Clinical Psychology Review 67, 22–35. doi:10.1016/j.cpr.2018.09.004.

Pompoli, A., Furukawa, T.A., Efthimiou, O., Imai, H., Tajika, A., Salanti, G., 2018. Dismantling cognitive-behaviour therapy for panic disorder: a systematic review and component network meta-analysis. Psychological Medicine 48, 1945–1953. doi:10.1017/s0033291717003919.

Preuss, T.M., Wise, S.P., 2022. Evolution of prefrontal cortex. Neuropsychopharmacology 47, 3–19. doi:10.1038/s41386-021-01076-5.

Quilty, L., McBride, C., Bagby, R., 2008. Evidence for the cognitive mediational model of cognitive behavioural therapy for depression. Psychological Medicine 38, 1531–1541. doi:10.1017/s0033291708003772.

Rahman, A., Malik, A., Sikander, S., Roberts, C., Creed, F., 2008. Cognitive behaviour therapy-based intervention by community health workers for mothers with depression and their infants in rural pakistan: a cluster-randomised controlled trial. The Lancet 372, 902–909. doi:10.1016/s0140-6736(08)61400-2.

Raio, C.M., Orederu, T.A., Palazzolo, L., Shurick, A.A., Phelps, E.A., 2013. Cognitive emotion regulation fails the stress test. Proceedings of the National Academy of Sciences 110, 15139–15144. doi:10.1073/pnas.1305706110.

Raio, C.M., Phelps, E.A., 2015. The influence of acute stress on the regulation of conditioned fear. Neurobiology of Stress 1, 134–146. doi:10.1016/j.ynstr.2014.11.004.

Rakovshik, S.G., McManus, F., Vazquez-Montes, M., Muse, K., Ougrin, D., 2016. Is supervision necessary? examining the effects of internet-based cbt training with and without supervision. Journal of Consulting and Clinical Psychology 84, 191–199. doi:10.1037/ccp0000079.

Richards, D.A., Ekers, D., McMillan, D., Taylor, R.S., Byford, S., Warren, F.C., Barrett, B., Farrand, P.A., Gilbody, S., Kuyken, W., O'Mahen, H., Watkins, E.R., Wright, K.A., Hollon, S.D., Reed, N., Rhodes, S., Fletcher, E., Finning, K., 2016. Cost and outcome of behavioural activation versus cognitive behavioural therapy for depression (cobra): a randomised, controlled, non-inferiority trial. The Lancet 388, 871–880. doi:10.1016/s0140-6736(16)31140-0.

Robinson, L., Delgadillo, J., Kellett, S., 2020. The dose-response effect in routinely delivered psychological therapies: A systematic review. Psychotherapy Research 30, 79–96. doi:10.1080/10503307.2019.1566676.

Roy, A., Hoge, E.A., Abrante, P., Druker, S., Liu, T., Brewer, J.A., 2021. Clinical efficacy and psychological mechanisms of an app-based digital

therapeutic for generalized anxiety disorder: Randomized controlled trial. Journal of Medical Internet Research 23, e26987. doi:10.2196/26987.

Roy-Byrne, P., Craske, M.G., Sullivan, G., Rose, R.D., Edlund, M.J., Lang, A.J., Bystritsky, A., Welch, S.S., Chavira, D.A., Golinelli, D., Campbell-Sills, L., Sherbourne, C.D., Stein, M.B., 2010. Delivery of evidence-based treatment for multiple anxiety disorders in primary care. JAMA 303, 1921. doi:10.1001/jama.2010.608.

Rozental, A., Magnusson, K., Boettcher, J., Andersson, G., Carlbring, P., 2017. For better or worse: An individual patient data meta-analysis of deterioration among participants receiving internet-based cognitive behavior therapy. Journal of Consulting and Clinical Psychology 85, 160–177. doi:10.1037/ccp0000158.

Safren, S.A., Otto, M.W., Sprich, S., Winett, C.L., Wilens, T.E., Biederman, J., 2005. Cognitive-behavioral therapy for adhd in medication-treated adults with continued symptoms. Behaviour Research and Therapy 43, 831–842. doi:10.1016/j.brat.2004.07.001.

Safren, S.A., Sprich, S., Mimiaga, M.J., Surman, C., Knouse, L., Groves, M., Otto, M.W., 2010. Cognitive behavioral therapy vs relaxation with educational support for medication-treated adults with adhd and persistent symptoms. JAMA 304, 875. doi:10.1001/jama.2010.1192.

Sandner, M., Zeier, P., Lois, G., Wessa, M., 2021. Cognitive emotion regulation withstands the stress test: An fmri study on the effect of acute stress on distraction and reappraisal. Neuropsychologia 157, 107876. doi:10.1016/j.neuropsychologia.2021.107876.

Saunders, R., Buckman, J.E., Stott, J., Leibowitz, J., Aguirre, E., John, A., Lewis, G., Cape, J., Pilling, S., 2021. Older adults respond better to psychological therapy than working-age adults: evidence from a large sample of mental health service attendees. Journal of Affective Disorders 294, 85–93. doi:10.1016/j.jad.2021.06.084.

Saxon, D., Barkham, M., 2012. Patterns of therapist variability: Therapist effects and the contribution of patient severity and risk. Journal of Consulting and Clinical Psychology 80, 535–546. doi:10.1037/a0028898.

Schermuly-Haupt, M.L., Linden, M., Rush, A.J., 2018. Unwanted events and side effects in cognitive behavior therapy. Cognitive Therapy and Research 42, 219–229. doi:10.1007/s10608-018-9904-y.

Schmeichel, B.J., Volokhov, R.N., Demaree, H.A., 2008. Working memory capacity and the self-regulation of emotional expression and experience. Journal of Personality and Social Psychology 95, 1526–1540. doi:10.1037/a0013345.

Schurr, M.S., Chen, Y.T., Raue, P.J., Areán, P.A., Alexopoulos, G.S., Renn, B.N., 2025. Relation between executive function test performance and treatment outcomes during brief psychotherapies for later-life depression. The American Journal of Geriatric Psychiatry: Open Science, Education, and Practice 7, 1–11. doi:10.1016/j.osep.2025.04.002.

Scott, M.J., 2018. Improving access to psychological therapies (iapt) - the need for radical reform. Journal of Health Psychology 23, 1136–1147. doi:10.1177/1359105318755264.

Shadish, W.R., Navarro, A.M., Matt, G.E., Phillips, G., 2000. The effects of psychological therapies under clinically representative conditions: A meta-analysis. Psychological Bulletin 126, 512–529. doi:10.1037/0033-2909.126.4.512.

Shafran, R., Clark, D., Fairburn, C., Arntz, A., Barlow, D., Ehlers, A., Freeston, M., Garety, P., Hollon, S., Ost, L., Salkovskis, P., Williams, J., Wilson, G., 2009. Mind the gap: Improving the dissemination of cbt. Behaviour Research and Therapy 47, 902–909. doi:10.1016/j.brat.2009.07.003.

Sharf, J., Primavera, L.H., Diener, M.J., 2010. Dropout and therapeutic alliance: A meta-analysis of adult individual psychotherapy. Psychotherapy: Theory, Research, Practice, Training 47, 637–645. doi:10.1037/a0021175.

Sheppes, G., Meiran, N., 2008. Divergent cognitive costs for online forms of reappraisal and distraction. Emotion 8, 870–874. doi:10.1037/a0013711.

Sheppes, G., Scheibe, S., Suri, G., Gross, J.J., 2011. Emotion-regulation choice. Psychological Science 22, 1391–1396. doi:10.1177/0956797611418350.

Shermohammed, M., Mehta, P.H., Zhang, J., Brandes, C.M., Chang, L.J., Somerville, L.H., 2017. Does psychosocial stress impact cognitive reappraisal? behavioral and neural evidence. Journal of Cognitive Neuroscience 29, 1803–1816. doi:10.1162/jocn_a_01157.

Shi, R., Sharpe, L., Abbott, M., 2019. A meta-analysis of the relationship between anxiety and attentional control. Clinical Psychology Review 72, 101754. doi:10.1016/j.cpr.2019.101754.

Shields, G.S., Sazma, M.A., Yonelinas, A.P., 2016. The effects of acute stress on core executive functions: A meta-analysis and comparison with cortisol. Neuroscience & Biobehavioral Reviews 68, 651–668. doi:10.1016/j.neubiorev.2016.06.038.

Solanto, M.V., Marks, D.J., Wasserstein, J., Mitchell, K., Abikoff, H., Alvir, J.M.J., Kofman, M.D., 2010. Efficacy of meta-cognitive therapy for adult adhd. American Journal of Psychiatry 167, 958–968. doi:10.1176/appi.ajp.2009.09081123.

Sonuga-Barke, E.J., Brandeis, D., Cortese, S., Daley, D., Ferrin, M., Holtmann, M., Stevenson, J., Danckaerts, M., van der Oord, S., Döpfner, M., et al., 2013. Nonpharmacological interventions for adhd: Systematic review and meta-analyses of randomized controlled trials of dietary and psychological treatments. American Journal of Psychiatry 170, 275–289. doi:10.1176/appi.ajp.2012.12070991.

Springer, K.S., Levy, H.C., Tolin, D.F., 2018. Remission in cbt for adult anxiety disorders: A meta-analysis. Clinical Psychology Review 61, 1–8. doi:10.1016/j.cpr.2018.03.002.

Steenkamp, M.M., Litz, B.T., Hoge, C.W., Marmar, C.R., 2015. Psychotherapy for military-related ptsd. JAMA 314, 489. doi:10.1001/jama.2015.8370.

Stewart, R.E., Chambless, D.L., 2009. Cognitive–behavioral therapy for adult anxiety disorders in clinical practice: A meta-analysis of effectiveness studies. Journal of Consulting and Clinical Psychology 77, 595–606. doi:10.1037/a0016032.

Stockton, D., Kellett, S., Berrios, R., Sirois, F., Wilkinson, N., Miles, G., 2019. Identifying the underlying mechanisms of change during acceptance

and commitment therapy (ACT): A systematic review of contemporary mediation studies. Behavioural and Cognitive Psychotherapy 47, 332–362. doi:10.1017/S1352465818000553.

Strunk, D.R., Brotman, M.A., DeRubeis, R.J., Hollon, S.D., 2010. Therapist competence in cognitive therapy for depression: Predicting subsequent symptom change. Journal of Consulting and Clinical Psychology 78, 429–437. doi:10.1037/a0019631.

Swift, J.K., Greenberg, R.P., 2012. Premature discontinuation in adult psychotherapy: A meta-analysis. Journal of Consulting and Clinical Psychology 80, 547–559. doi:10.1037/a0028226.

Tackett, J.L., Brandes, C.M., King, K.M., Markon, K.E., 2019. Psychology's replication crisis and clinical psychological science. Annual Review of Clinical Psychology 15, 579–604. doi:10.1146/annurev-clinpsy-050718-095710.

Tang, T.Z., DeRubeis, R.J., 1999. Sudden gains and critical sessions in cognitive-behavioral therapy for depression. Journal of Consulting and Clinical Psychology 67, 894–904. doi:10.1037/0022-006x.67.6.894.

Tang, T.Z., DeRubeis, R.J., Beberman, R., Pham, T., 2005. Cognitive changes, critical sessions, and sudden gains in cognitive-behavioral therapy for depression. Journal of Consulting and Clinical Psychology 73, 168–172. doi:10.1037/0022-006x.73.1.168.

Tapp, K., Vereenooghe, L., Hewitt, O., Scripps, E., Gray, K.M., Langdon, P.E., 2023. Psychological therapies for people with intellectual disabilities: An updated systematic review and meta-analysis. Comprehensive Psychiatry 122, 152372. doi:10.1016/j.comppsych.2023.152372.

Taylor, S., Abramowitz, J.S., McKay, D., 2012. Non-adherence and non-response in the treatment of anxiety disorders. Journal of Anxiety Disorders 26, 583–589. doi:10.1016/j.janxdis.2012.02.010.

Taylor, V.A., Moseley, I., Sun, S., Smith, R., Roy, A., Ludwig, V.U., Brewer, J.A., 2021. Awareness drives changes in reward value which predict eating behavior change: Probing reinforcement learning using experience sampling from mobile mindfulness training for maladaptive eating. Journal of Behavioral Addictions 10, 482–497. doi:10.1556/2006.2021.00020.

Tolin, D.F., 2010. Is cognitive–behavioral therapy more effective than other therapies? Clinical Psychology Review 30, 710–720. doi:10.1016/j.cpr.2010.05.003.

Topper, M., Emmelkamp, P.M., Watkins, E., Ehring, T., 2017. Prevention of anxiety disorders and depression by targeting excessive worry and rumination in adolescents and young adults: A randomized controlled trial. Behaviour Research and Therapy 90, 123–136. doi:10.1016/j.brat.2016.12.015.

Truijens, F., Zühlke-van Hulzen, L., Vanheule, S., 2019. To manualize, or not to manualize: Is that still the question? a systematic review of empirical evidence for manual superiority in psychological treatment. Journal of Clinical Psychology 75, 329–343. doi:10.1002/jclp.22712.

Vandborg, S.K., Hartmann, T.B., Bennedsen, B.E., Pedersen, A.D., Thomsen, P.H., 2016. Can memory and executive functions in patients with obsessive–compulsive disorder predict outcome of cognitive behavioural therapy? Nordic Journal of Psychiatry 70, 183–189. doi:10.3109/08039488.2015.1074282.

Vervliet, B., Craske, M.G., Hermans, D., 2013. Fear extinction and relapse: State of the art. Annual Review of Clinical Psychology 9, 215–248. doi:10.1146/annurev-clinpsy-050212-185542.

Vittengl, J.R., Clark, L.A., Dunn, T.W., Jarrett, R.B., 2007. Reducing relapse and recurrence in unipolar depression: A comparative meta-analysis of cognitive-behavioral therapy's effects. Journal of Consulting and Clinical Psychology 75, 475–488. doi:10.1037/0022-006x.75.3.475.

Vittengl, J.R., Clark, L.A., Thase, M.E., Jarrett, R.B., 2014. Are improvements in cognitive content and depressive symptoms correlates or mediators during acute-phase cognitive therapy for recurrent major depressive disorder? International Journal of Cognitive Therapy 7, 251–271. doi:10.1521/ijct.2014.7.3.251.

Wakefield, S., Kellett, S., Simmonds-Buckley, M., Stockton, D., Bradbury, A., Delgadillo, J., 2021. Improving access to psychological therapies (iapt) in the united kingdom: A systematic review and meta-analysis of 10-years

of practice-based evidence. British Journal of Clinical Psychology 60, 1–37. doi:10.1111/bjc.12259.

Waller, G., 2009. Evidence-based treatment and therapist drift. Behaviour Research and Therapy 47, 119–127. doi:10.1016/j.brat.2008.10.018.

Waller, G., Turner, H., 2016. Therapist drift redux: Why well-meaning clinicians fail to deliver evidence-based therapy, and how to get back on track. Behaviour Research and Therapy 77, 129–137. doi:10.1016/j.brat.2015.12.005.

Wampold, B.E., 2015. How important are the common factors in psychotherapy? an update. World Psychiatry 14, 270–277. doi:10.1002/wps.20238.

Wang, X., He, H., Lan, J., Guo, Y., Zhang, Y., 2026. Cognitive behavioral therapy for anxiety symptoms in adhd: A systematic review and meta-analysis with dose–response exploration. Psychotherapy Research , 1–19doi:10.1080/10503307.2026.2658745.

Watkins, E.R., Mullan, E., Wingrove, J., Rimes, K., Steiner, H., Bathurst, N., Eastman, R., Scott, J., 2011. Rumination-focused cognitive–behavioural therapy for residual depression: phase ii randomised controlled trial. British Journal of Psychiatry 199, 317–322. doi:10.1192/bjp.bp.110.090282.

Webb, C.A., DeRubeis, R.J., Barber, J.P., 2010. Therapist adherence/competence and treatment outcome: A meta-analytic review. Journal of Consulting and Clinical Psychology 78, 200–211. doi:10.1037/a0018912.

Weisman, J.S., Rodebaugh, T.L., 2018. Exposure therapy augmentation: A review and extension of techniques informed by an inhibitory learning approach. Clinical Psychology Review 59, 41–51. doi:10.1016/j.cpr.2017.10.010.

Weiss, M., Murray, C., Wasdell, M., Greenfield, B., Giles, L., Hechtman, L., 2012. A randomized controlled trial of cbt therapy for adults with adhd with and without medication. BMC Psychiatry 12, 30. doi:10.1186/1471-244x-12-30.

Weisz, J.R., Kuppens, S., Eckshtain, D., Ugueto, A.M., Hawley, K.M., Jensen-Doss, A., 2013. Performance of evidence-based youth psychotherapies

compared with usual clinical care. JAMA Psychiatry 70, 750. doi:10.1001/jamapsychiatry.2013.1176.

Weitz, E.S., Hollon, S.D., Twisk, J., van Straten, A., Huibers, M.J.H., David, D., et al., 2015. Baseline depression severity as moderator of depression outcomes between cognitive behavioral therapy vs pharmacotherapy. JAMA Psychiatry 72, 1102. doi:10.1001/jamapsychiatry.2015.1516.

Wenzlaff, R.M., Wegner, D.M., 2000. Thought suppression. Annual Review of Psychology 51, 59–91. doi:10.1146/annurev.psych.51.1.59.

Weston, L., Hodgekins, J., Langdon, P.E., 2016. Effectiveness of cognitive behavioural therapy with people who have autistic spectrum disorders: A systematic review and meta-analysis. Clinical Psychology Review 49, 41–54. doi:10.1016/j.cpr.2016.08.001.

Wiles, N., Thomas, L., Abel, A., Ridgway, N., Turner, N., Campbell, J., Garland, A., Hollinghurst, S., Jerrom, B., Kessler, D., Kuyken, W., Morrison, J., Turner, K., Williams, C., Peters, T., Lewis, G., 2013. Cognitive behavioural therapy as an adjunct to pharmacotherapy for primary care based patients with treatment resistant depression: results of the cobalt randomised controlled trial. The Lancet 381, 375–384. doi:10.1016/s0140-6736(12)61552-9.

Wiles, N.J., Thomas, L., Turner, N., Garfield, K., Kounali, D., Campbell, J., Kessler, D., Kuyken, W., Lewis, G., Morrison, J., Williams, C., Peters, T.J., Hollinghurst, S., 2016. Long-term effectiveness and cost-effectiveness of cognitive behavioural therapy as an adjunct to pharmacotherapy for treatment-resistant depression in primary care: follow-up of the cobalt randomised controlled trial. The Lancet Psychiatry 3, 137–144. doi:10.1016/s2215-0366(15)00495-2.

Willcutt, E.G., Doyle, A.E., Nigg, J.T., Faraone, S.V., Pennington, B.F., 2005. Validity of the executive function theory of attention-deficit/hyperactivity disorder: A meta-analytic review. Biological Psychiatry 57, 1336–1346. doi:10.1016/j.biopsych.2005.02.006.

Williams, A.C.d.C., Fisher, E., Hearn, L., Eccleston, C., 2020. Psychological therapies for the management of chronic pain (excluding headache) in

adults. Cochrane Database of Systematic Reviews 2021. doi:10.1002/14651858.cd007407.pub4.

de Wit, S., Kindt, M., Knot, S.L., Verhoeven, A.A.C., Robbins, T.W., Gasull-Camos, J., Evans, M., Mirza, H., Gillan, C.M., 2018. Shifting the balance between goals and habits: Five failures in experimental habit induction. Journal of Experimental Psychology: General 147, 1043–1065. doi:10.1037/xge0000402.

Wolgast, M., 2014. What does the acceptance and action questionnaire (aaq-ii) really measure? Behavior Therapy 45, 831–839. doi:10.1016/j.beth.2014.07.002.

Wood, W., Rünger, D., 2016. Psychology of habit. Annual Review of Psychology 67, 289–314. doi:10.1146/annurev-psych-122414-033417.

Wu, J.Q., Appleman, E.R., Salazar, R.D., Ong, J.C., 2015. Cognitive behavioral therapy for insomnia comorbid with psychiatric and medical conditions. JAMA Internal Medicine 175, 1461. doi:10.1001/jamainternmed.2015.3006.

Young, S., Emilsson, B., Sigurdsson, J.F., Khondoker, M., Philipp-Wiegmann, F., Baldursson, G., Olafsdottir, H., Gudjonsson, G., 2017. A randomized controlled trial reporting functional outcomes of cognitive–behavioural therapy in medication-treated adults with adhd and comorbid psychopathology. European Archives of Psychiatry and Clinical Neuroscience 267, 267–276. doi:10.1007/s00406-016-0735-0.

Young, Z., Moghaddam, N., Tickle, A., 2020. The efficacy of cognitive behavioral therapy for adults with adhd: A systematic review and meta-analysis of randomized controlled trials. Journal of Attention Disorders 24, 875–888. doi:10.1177/1087054716664413.

Öst, L.G., 2008. Efficacy of the third wave of behavioral therapies: A systematic review and meta-analysis. Behaviour Research and Therapy 46, 296–321. doi:10.1016/j.brat.2007.12.005.

Öst, L.G., 2014. The efficacy of acceptance and commitment therapy: An updated systematic review and meta-analysis. Behaviour Research and Therapy 61, 105–121. doi:10.1016/j.brat.2014.07.018.

Šipka, D., Lopes, R., Krieger, T., Klein, J.P., Berger, T., 2025. Active components in internet-based cognitive-behavioral therapy for social anxiety disorder: A randomized full factorial trial. Psychotherapy and Psychosomatics , 1–20doi:10.1159/000542425.